

Post-Hurricane Habitat Use of the Puerto Rican Emerald along Elevation and Vegetation Greenness Gradients and Implications for Monitoring

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Abstract

Bird occurrence and occupancy changes across environments after hurricanes inform monitoring priorities. We analyze 89,682 eBird checklists for the Puerto Rican Emerald (*Riccordia maugaeus*), 2013–2025, combining elevation and vegetation-greenness stratification, dynamic occupancy modeling, and spatial and temporal holdout prediction to examine habitat-use differences and environmental coverage under alternative record-screening policies. Overall reporting rates were 7.59%, 4.62% and 6.21% before María, between hurricanes and after Fiona. Standardized predictions retained high reporting levels and larger absolute rebound point estimates in high-elevation, high-greenness environments. Some lowland predictions remained below their initial levels, although all stratum-specific final-versus-initial contrast intervals spanned zero. In the corrected dynamic model, mean occupancy persistence probability was 96.8% in high-elevation, high-greenness grids and 24.2% and 71.3% in low- and high-greenness lowland grids. This is the probability that a previously occupied grid remains occupied in the next primary period. Greenness was more clearly associated with occupancy persistence than with colonization, indicating that low aggregate lowland reporting can coexist with higher persistence in greener grids. Prediction gains concentrated in identification of sparse low- and middle-elevation reports. Under equal checklist-screening budgets, elevation-stratified ranking increased lowland positive-report coverage from 34.6% to 45.6%, while reducing overall coverage from 46.7% to 35.2%. Prioritizing only high reporting scores therefore does not provide balanced environmental coverage. Monitoring should retain fixed upland revisits, cover contrasting lowland greenness conditions, and use within-stratum ranking to identify supplementary candidates. Later repeat reports should be recorded separately from occupancy persistence and colonization estimated under imperfect detection, allowing environmental differences and their temporal changes to be assessed.

Keywords: Puerto Rican Emerald; hurricane-related change; eBird; vegetation greenness; occupancy persistence; dynamic occupancy; spatial validation; reanalysis

1 Introduction

Tropical cyclones can alter forest structure and bird habitat over time scales extending well beyond the storm itself. Research in Puerto Rico has documented both forest damage and subsequent bird responses, providing a basis for asking how disturbance-related change varies across an island landscape [1, 2]. Recent work on the Puerto Rico Plain Pigeon documents a prolonged post-hurricane population bottleneck after accounting for detection, while remote sensing of Puerto Rican mangroves shows that canopy damage and vegetation-index recovery vary among sites [3, 4]. These studies motivate environmental comparisons without assuming that vegetation recovery and bird recovery follow the same trajectory. For an endemic resident species, a rise in records after a period of low reporting needs to be examined in its environmental context. Analysis must locate the increase, test whether elevation and vegetation greenness are associated with previously occupied sites remaining occupied in the next period, and assess whether the relationships can

guide monitoring elsewhere. Addressing these questions requires keeping the observed pattern, its ecological interpretation and its predictive usefulness distinct.

The Puerto Rican Emerald (*Riccordia maugaeus*) offers an opportunity to examine this sequence. Ortiz-Andrade and Rojas-Perez [5] combined eBird checklists, elevation and satellite vegetation greenness to describe three periods defined relative to Hurricanes María and Fiona. Their checklist-level models linked reporting to environmental gradients, while a dynamic occupancy model addressed occupancy persistence under imperfect detection. Their public reproducibility package permits a reanalysis of these complementary lines of evidence. It also permits a distinction between reproducing a reported number and checking whether the implementation represents the intended statistical model.

The first interpretive challenge concerns the response being measured. An eBird checklist describes a reporting event, whose probability depends on species presence, survey effort and other observation conditions [6, 7]. Backstrom et al. [8] further show that spatial and temporal sampling biases can interact in eBird data and alter trend estimates, making the environmental composition of the sample relevant to period comparisons. A change in the proportion of positive checklists is therefore not, by itself, a change in population size. Elevation-specific predictions can reveal differences hidden by island-wide averages, but they remain conditional reporting probabilities. Dynamic occupancy models estimate latent site occupancy and transitions between occupied and unoccupied states, conditional on repeat-survey and detection assumptions [9–11]. A recent Prairie Warbler study used long-term point counts and dynamic occupancy models to distinguish colonization and local extinction responses to habitat management [12], illustrating why site retention and entry into occupancy should be evaluated separately. These quantities must be interpreted on their own terms. In particular, a high occurrence score or its complement does not measure a local extinction transition.

The second challenge concerns the reliability of the observation process used to infer those states. Environmental associations with persistence are informative only insofar as observations and covariates are correctly aligned. A reproducible script may reproduce an unintended arrangement just as consistently as an intended one. Examining implementation errors helps assess the reliability of ecological interpretations. A correction may retain the environmental association while changing the evidence for a detection difference, making separate evaluation of these explanations necessary.

The third challenge is geographic prediction. A relationship that summarizes the sampled locations may not retain the same value in held-out geographic blocks. Repeated visits and geographically clustered checklists make the validation design relevant to the intended monitoring application [13–15]. Random and spatial partitions ask different questions, and good performance under one does not establish performance under the other. Recent comparisons of species-distribution models confirm that random validation can overestimate transfer performance [16]; tests of spatial versus temporal environmental relationships also show that transferability can vary within a species’ range [17]. Moreover, predicting checklist reports is not an independent validation of a latent occupancy model. Ecological explanation and reporting prediction contribute different evidence to a monitoring decision [18]. In bird forecasting, Ovaskainen et al. [19] combined citizen-science recordings with standardized intervals and permanent point-count networks, illustrating how prediction can be supported by a repeatable observation design.

We examine reporting rebound across elevation and greenness conditions, test whether these environments also have higher occupancy persistence probabilities, and use the results to assess how environmental differences should be monitored. First, missingness composition and standardized reporting predictions separate environmental representation from within-stratum period change. Second, corrected period-label alignment in the dynamic occupancy model allows reporting trajectories to be compared with occupancy persistence probability, colonization and repeat reports in corresponding environmental cells. Finally, spatial and temporal validation and equal-budget stratified ranking assess whether reporting predictions can supply clues for revisits in target environments. Conventional benchmarks and checks of the exploratory neural workflow serve this monitoring question. As a secondary analysis of the same public observations, the study connects the environmental distribution of rebound, corrected persistence associations and practical tradeoffs

in monitoring coverage.

2 Materials and Methods

2.1 Study design, observations and environmental coverage

We used the derived data and scripts released by Ortiz-Andrade and Rojas-Perez [5], together with the existing experiment outputs in this project. The study concerns Puerto Rico and retains complete eBird checklists using Stationary or Traveling protocols, durations of 5–240 min and travel distances no greater than 5 km. The response is one if the species was reported and zero otherwise. Zero denotes a non-report, not verified absence.

Habitat refers to the environment in which a species lives. We characterize selected habitat conditions through elevation and vegetation greenness and examine their associations with checklist records and modeled occupancy states. Habitat use is a broad description of these associations here; foraging, breeding behavior and habitat quality were not measured directly. At the observation level, a repeat report means that a checklist reports the species again at the same site in a later period. It depends on both presence and detection and is reported separately from modeled occupancy persistence.

The analysis covers 1 January 2013 to 31 December 2025. We retain the published period definitions. Pre-María ends on 19 September 2017; Inter-Hurricanes runs from 20 September 2017 to 17 September 2022; and Post-Fiona begins on 18 September 2022. These categories locate observations relative to the storms. They do not measure local storm exposure or supply an undisturbed counterfactual, so their contrasts are temporal associations rather than estimates of hurricane causation.

Elevation derives from SRTM [20]; NDVI summarizes vegetation greenness [21]. The released Landsat procedure selects the nearest valid observation within ± 16 days and restricts imagery to the relevant period. We use the released covariates rather than repeating satellite extraction. The full analysis contains 89,682 checklists, of which 68,976 have NDVI; 20,706 (23.1%) do not. Environmental GLMs and the principal conventional benchmark use the complete subset. We describe missingness by period and reporting outcome without assuming it is random. This composition check is motivated by evidence that spatial and temporal sampling biases can interact in community-science trend estimation [8]. The archived EVI2 comparison supplies a secondary vegetation-index check [22].

Figure 1 places checklist coverage and the validation blocks in the same landscape. Its 24,876 locations refer to the pre-extraction coordinate table; the exported environmental table contains 24,856 coordinate pairs because coordinate precision differs. Neither count is the number of occupancy sites or validation blocks. The sampling locations reflect opportunities for observation rather than a spatially uniform survey.

2.2 Verification and conditional reporting models

File hashes, sample counts, filtering criteria and fitted coefficients were checked for data and model reproduction. Two duplicated date–coordinate records were retained to preserve the original analysis sample. Table 1 summarizes source selection. The reference study scored 5.1 for implementation suitability, compared with 3.4–4.1 for the other candidates. Its combination of observations, environmental covariates, analysis scripts and checksums enabled a linked evaluation of reporting patterns, occupancy processes and predictive validation from a common input. Data verification passed 53 of 54 checks; all 44 GLM checks met their tolerances, and all 11 occupancy coefficients and standard errors were reproduced (Figure 2; Table 2). The sole data-check discrepancy involved two excess records with duplicated date–coordinate combinations, approximately 0.0022% of all checklists, both without a species report. This localized the subsequent investigation to observation-covariate alignment within the model rather than differences in input versions or filtering.

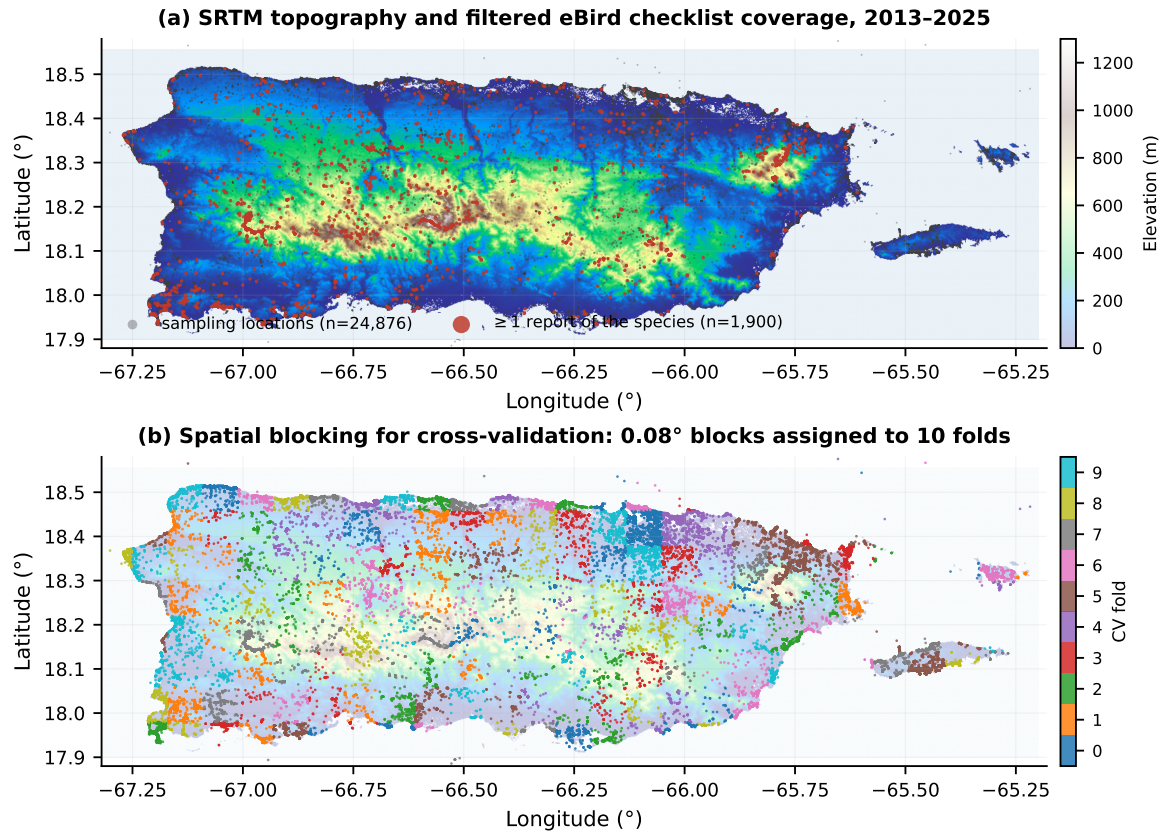


Figure 1: Puerto Rico, SRTM elevation and checklist coverage (upper panel), with one spatial fold allocation (lower panel). Entire 0.08° blocks are assigned to folds; adjacent training and test blocks have no distance buffer.

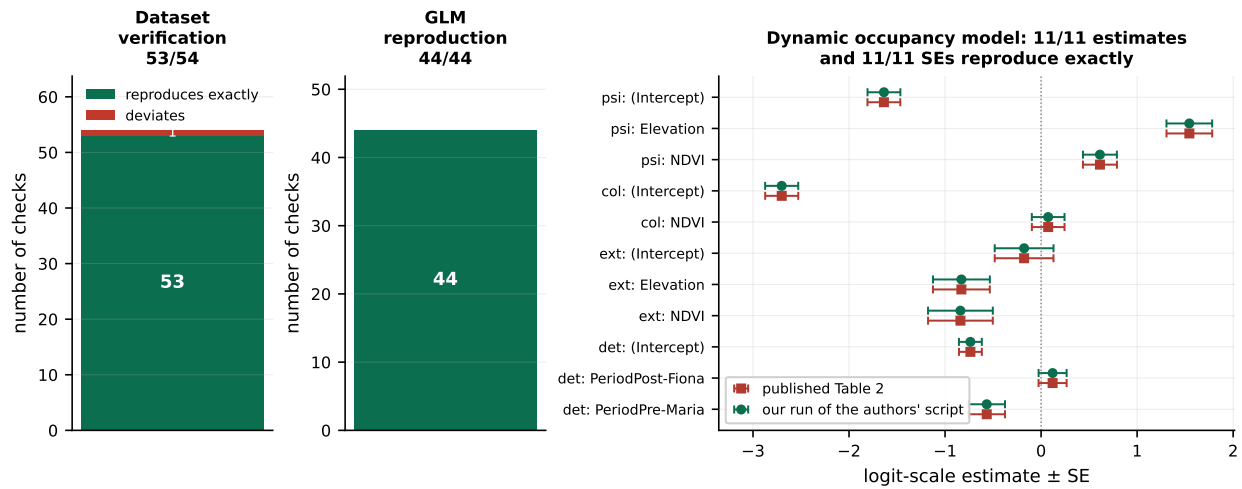


Figure 2: Data verification and numerical model reproduction. Colors distinguish check categories; coefficient comparisons use the original model ordering.

Table 1: Source selection and implementation suitability. Scores weight data accessibility, time requirements, implementation convenience and figure resources by 0.4, 0.3, 0.2 and 0.1.

#	DOI suffix	Title (truncated)	Data availability	Score
1	birds7030055	Elevation and Vegetation Greenness Structure Post-Hurricane Resilience and Loc	public (MDPI supplement: full reproducibility package, SHA	5.1
2	birds7030045	Variation in Avian Predation Potential on Insect Pests Across Time and Space W	public (Mendeley Data, HTTP 200; data + R code)	4.1
3	birds7030056	Urbanization Erodes Plant-Hummingbird Networks Mainly Through Interaction Rewi	public (Mendeley Data, HTTP 200)	4.1
4	birds7010019	Migratory Status Shapes Exploratory Behavior but Not Learning Performance in H	public (Mendeley Data, HTTP 200)	4.0
5	birds7020025	Death Cause of Eastern Red-Rumped Swallows (<i>Cecropis daurica</i>) in a Mass-Mortal	public (Zenodo, HTTP 200)	4.0
6	birds7030049	Incubation Behavior of Threatened <i>Oreopholus ruficollis</i> in the Atacama Arid Ec	public (GitHub, HTTP 200)	3.9
7	birds7010016	Correlates of Bird Visits to One Single Tree in Durban, South Africa: Ecologic	Supplementary xlsx	3.8
8	birds7010011	Multivariate Assessment of Geographic and Ecological Drivers of Heavy Metal Ac	in article + Supplementary xlsx (Tables S1-)	3.8
9	birds7030041	Slow Spread of the Introduced Oriental Magpie in Dispersed Urban Habitats on H	in article + Supplementary (source_data.zip, tracks&magpie	3.7
10	birds7010017	Anthropogenic and Environmental Factors Influence Kentish Plover (<i>Anarhynchus</i>	public (Dryad, HTTP 200) for recovery data; biometrics on	3.4

Table 2: All 54 archived dataset checks, including the date-coordinate duplicate exception. VT denotes the released validation targets. Full-export counts and 2013–2025 analysis counts have different denominators.

Check	Reported	Recomputed	Match
[VT] Filtered pre-GEE table: total rows	91677	91677	yes
[VT] Filtered pre-GEE table: rows 2013-2025	89682	89682	yes
[VT] Filtered pre-GEE table: detections (full)	5326	5326	yes
[VT] Filtered pre-GEE table: non-detections (full)	86351	86351	yes
[VT] Corrected NDVI export: total rows	91677	91677	yes
[VT] Corrected NDVI export: rows 2013-2025	89682	89682	yes
[VT] Corrected NDVI export: detections (full)	5326	5326	yes
[VT] Corrected NDVI export: non-detections (full)	86351	86351	yes
[VT] Corrected NDVI export: non-missing index (full)	70799	70799	yes
[VT] EVI2 export: total rows	91677	91677	yes
[VT] EVI2 export: rows 2013-2025	89682	89682	yes
[VT] EVI2 export: detections (full)	5326	5326	yes
[VT] EVI2 export: non-detections (full)	86351	86351	yes
[VT] EVI2 export: non-missing index (full)	70799	70799	yes
paper 3.1: checklists analysed 2013-2025	89682	89682	yes
paper 3.1: checklists reporting species	5174	5174	yes
paper 3.1: checklists not reporting	84508	84508	yes
paper 3.1: overall encounter frequency (%)	5.77	5.77	yes
Table 1 Pre-Maria: total checklists	14728	14728	yes
Table 1 Pre-Maria: reporting species	1118	1118	yes
Table 1 Pre-Maria: encounter frequency (%)	7.59	7.59	yes
Table 1 Inter-Hurricanes: total checklists	37701	37701	yes
Table 1 Inter-Hurricanes: reporting species	1742	1742	yes

Check	Reported	Recomputed	Match
Table 1 Inter-Hurricanes: encounter frequency (%)	4.62	4.62	yes
Table 1 Post-Fiona: total checklists	37253	37253	yes
Table 1 Post-Fiona: reporting species	2314	2314	yes
Table 1 Post-Fiona: encounter frequency (%)	6.21	6.21	yes
paper 3.1: Pre-Maria share (%)	16.4	16.4	yes
paper 3.1: Inter-Hurricanes share (%)	42.0	42.0	yes
paper 3.1: Post-Fiona share (%)	41.5	41.5	yes
paper 3.1: checklists with missing NDVI	20706	20706	yes
paper 3.1: missing-NDVI share (%)	23.1	23.1	yes
paper 2.4.2/3.1: NDVI-complete analytical subset	68976	68976	yes
EVI2 subset size (parity with NDVI)	68976	68976	yes
paper 2.1/Fig1: unique sampling locations	24876	24876	yes
paper 2.1/Fig1: unique locations with ≥ 1 report	1900	1900	yes
integrity: unique sampling-event IDs == rows (full)	91677	91677	yes
integrity: fully duplicated rows (full)	0	0	yes
integrity: duplicated (date,lon,lat) triples	0	2	no
integrity: missing values in core covariates	0	0	yes
integrity: Presence coded 0,1 only	True	True	yes
integrity: Duration within 5-240 min filter	True	True	yes
integrity: Distance ≤ 5 km filter	True	True	yes
integrity: Elevation_m ≥ 0 (negatives set to 0)	True	True	yes
integrity: NDVI within [-1,1]	True	True	yes
integrity: Days_from_eBird < 17 (fractional days)	True	True	yes
integrity: NDVI/EVI2 exports row-aligned on IDs	True	True	yes
paper 2.7: initially eligible sites (≥ 1 checklist per period)	1048	1048	yes
paper 2.7: sites dropped for no valid NDVI	4	4	yes
paper 2.7: final occupancy sites	1044	1044	yes
paper 2.7: observed survey occasions	7908	7908	yes
paper 2.7: missing occasions in 1044x9 matrix	1488	1488	yes
paper 3.5: Inter-Hurricanes time-since-Maria N	28707	28707	yes
paper 3.5: Post-Fiona time-since-Fiona N	29580	29580	yes

Five binomial GLMs progress from an intercept through period, observation effort, environment and period–elevation interaction. All use the same NDVI-complete subset. The published selected model is

$$\text{logit}(q_i) = \alpha + \beta_{P(i)} + \beta_E E_i + \delta_{P(i)} E_i + \beta_V V_i + \beta_D D_i + \beta_L L_i, \quad (1)$$

where q_i is reporting probability, E_i elevation in 100-m units, V_i NDVI, D_i duration in minutes and L_i distance in kilometres. Pre-Maria is the reference period. We report existing AIC, model weights and pseudo- R^2 as summaries of the candidate fits [23, 24], separately from held-out performance.

For the elevation curves, NDVI, duration and distance are fixed at their complete-sample means. Predictions at 100 and 900 m summarize low and high elevations, not biological thresholds. Following the released analysis, we also summarize the curve on a uniform 0–1300 m grid at 10-m intervals:

$$C_t = \frac{\sum_h h \hat{q}_t(h)}{\sum_h \hat{q}_t(h)}. \quad (2)$$

This probability-weighted elevation centroid describes a fitted curve, not population abundance, geographic range centre or tracked movement. Annual reporting intervals reproduce the archived binomial intervals and do not account for observer or geographic dependence.

2.3 Occupancy persistence and observation-model correction

The dynamic model uses 0.01° rounded-coordinate cells as sites and the three periods as primary occasions. Sites require a checklist in every period. Within each site–period, the first three records after ordering by year and original row order are retained, with fewer visits represented by missing cells. The resulting 1,044-by-nine detection history contains 7,908 observed occasions and 1,488 missing positions. Site covariates are computed from all eligible records before visit selection and

standardized across retained sites. NDVI is therefore a cross-time site summary, not a within-site vegetation-recovery trajectory.

The selected **unmarked** model [25] relates initial occupancy (ψ) and local extinction (ε) to elevation and NDVI, colonization (γ) to NDVI, and detection (p) to period. Period does not separately enter the transition components. Thus the model does not estimate a storm-specific change in extinction or colonization.

Let $z_{g,t}$ denote the latent occupancy state of grid g in primary period t . Occupancy persistence probability is defined as $\Pr(z_{g,t+1} = 1 \mid z_{g,t} = 1) = 1 - \varepsilon_g$, the probability that a previously occupied grid remains occupied in the next period [10]. It does not represent residence of the same individual, individual survival or uninterrupted presence throughout a period. Colonization probability γ conditions on prior nonoccupancy, whereas an observed first or repeat report does not directly identify colonization. Persistence throughout this paper refers to this grid-level state transition.

The observation covariate must align with the detection matrix in site-major order. All nine observations from one site precede those of the next. The released script instead repeats each occasion label across sites. The archived labelled-frame check and controlled refit change only this alignment, keeping the response, site covariates, formula and optimization settings fixed. We compare estimates and uncertainty, using $\hat{\beta} \pm 1.96 \text{ SE}$ for approximate 95% Wald intervals. This is a sensitivity test within the selected model, not a new selection across all corrected candidate models. Multi-year primary periods also place substantial demands on within-period closure; estimated transitions are not annual demographic rates.

2.4 Reporting prediction in held-out geographic blocks

The prediction target remains a checklist report. The eleven conventional configurations are GLM m3, GLM m4, a spline GLM, m4 with terrain, random forest, LightGBM, XGBoost, a multilayer perceptron (MLP), and terrain-augmented variants of the last three. Tree methods provide flexible tabular predictors [26–28]. Their features include environment, effort, period, seasonal terms and coordinates; the published GLMs retain their original covariates. Terrain configurations add 20 SRTM descriptors of elevation, slope, aspect and neighbourhood variation at four scales. Consequently, comparisons concern implemented model–information combinations, not isolated architecture effects.

The saved settings use 400 trees and a minimum leaf size of five for random forest; the boosting implementations use 600 estimators and learning rate 0.05, with 63 leaves for LightGBM and maximum depth six for XGBoost. The MLP has three hidden layers of width 128. Estimated transformations, including MLP scaling and spline knots, are fitted using each training fold. These settings describe the existing runs; we do not retune against the reported tests.

The primary protocol assigns entire 0.08° geographic blocks, approximately 8–9 km per side, to ten size-balanced folds, repeated with three block origins. The eleven configurations share all 30 test folds. Each fold can contain disconnected blocks; no buffer separates adjoining training and test blocks. The protocol tests the specified within-island geographic holdout, not arbitrary-distance extrapolation. Secondary random validation uses three stratified five-fold repetitions. Training fractions and test composition follow the respective partition rules. The archived temporal experiment repeats fits on the same Post-Fiona test sample with three seeds; it evaluates transfer of relationships learned in earlier periods to later reports. Separate geographic and temporal holdouts follow the principle that validation should represent the intended transfer task [16].

We emphasize AUC, average precision (AP), Brier score and recall among the highest-ranked 10% of test checklists [29, 30]. They address overall discrimination, positive-report ranking, probability error and a fixed ranking budget, respectively. Comparisons use unweighted fold means and between-fold standard deviations, absolute differences, relative improvements and top-decile report coverage. Training sets overlap across repeated partitions, so folds are not treated as independent replicates for significance classification. The pooled curves use a separate ten-fold prediction export. ROC and precision–recall characterize ranking, while reliability diagrams use 12 equal-count bins

and expected calibration error uses 15. Scale sensitivity evaluates 0.04° , 0.08° , 0.16° and 0.32° blocks; the EVI2 substitution uses six folds.

2.5 Environmental stratification, process probabilities and monitoring budgets

Supplementary analyses stratified elevation as below 300 m, 300–below 600 m, and at least 600 m. NDVI tertile cutpoints of 0.4172 and 0.7006, computed from 68,976 complete checklists, produced nine environmental combinations. These exploratory strata describe heterogeneity rather than species habitat thresholds. Occupancy summaries apply the same NDVI cutpoints to cross-period site means, report site counts, and flag combinations with fewer than ten sites. NDVI-missing comparisons first summarize report and missingness rates within elevation bands. Binomial reporting models then relate reports to a missingness indicator, sequentially adding period and an elevation spline, followed by log duration, $\log(1 + x)$ distance, quarter, and quadratic geographic-coordinate terms. Standard errors cluster 214 geographic blocks of width 0.08° . Trust-region optimization supplies stable starting values, with convergence, gradient and covariance checks. Reporting rebound is standardized using a refit of the reproduced m4 formula. Each environmental cell retains its pooled covariate distribution while period is replaced in turn. Post-Fiona minus Pre-María intervals use clustered covariance and the delta method; raw stratified rates additionally use 1,000 spatial block bootstrap samples. The corrected occupancy model was refitted on the DGX, reproducing existing corrected coefficients within 10^{-4} . Site predictions of initial occupancy, occupancy persistence probability $1 - \varepsilon$, and colonization γ were averaged by environmental cell. Two thousand joint normal coefficient draws preserve the full covariance and yield 95% probability and contrast intervals. A contrast at 100 m from NDVI 0.30 to 0.76 isolates the fitted greenness response; these values approximate the low- and high-greenness medians among low-elevation sites. Adjacent period pairs with three visits in each period were also extracted from the nine-column detection history to summarize repeated reports conditional on a previous report. This equal-visit reporting proportion is presented separately from latent occupancy transitions. A new random forest used the original 400 trees, minimum leaf size five and seed zero in one ten-fold spatial partition. Every fold was checked for disjoint training/test event identifiers and geographic blocks; recomputing the first m4 fold validated the row alignment of the saved predictions. Stratified comparisons include m4, random forest and both terrain-augmented boosting models. Each test fold receives a budget equal to the floor of 10% of its checklist count. Global ranking selects the highest scores within that fold; stratified ranking allocates approximately 10% within each elevation band, using largest remainders to preserve exactly the same fold budget. Both select 6,890 checklists. AP, Brier, selection fractions and positive-report coverage are reported by stratum. Coverage-contrast intervals use 1,000 spatial block bootstrap samples conditional on this partition’s fitted predictions. The experiment evaluates record ranking and environmental coverage; checklist counts are not converted into field-visit costs.

Figure 3 summarizes how the environmental and survey data support complementary analyses of conditional reporting, occupancy persistence and reporting prediction, and how these analyses inform follow-up monitoring across elevation and greenness conditions.

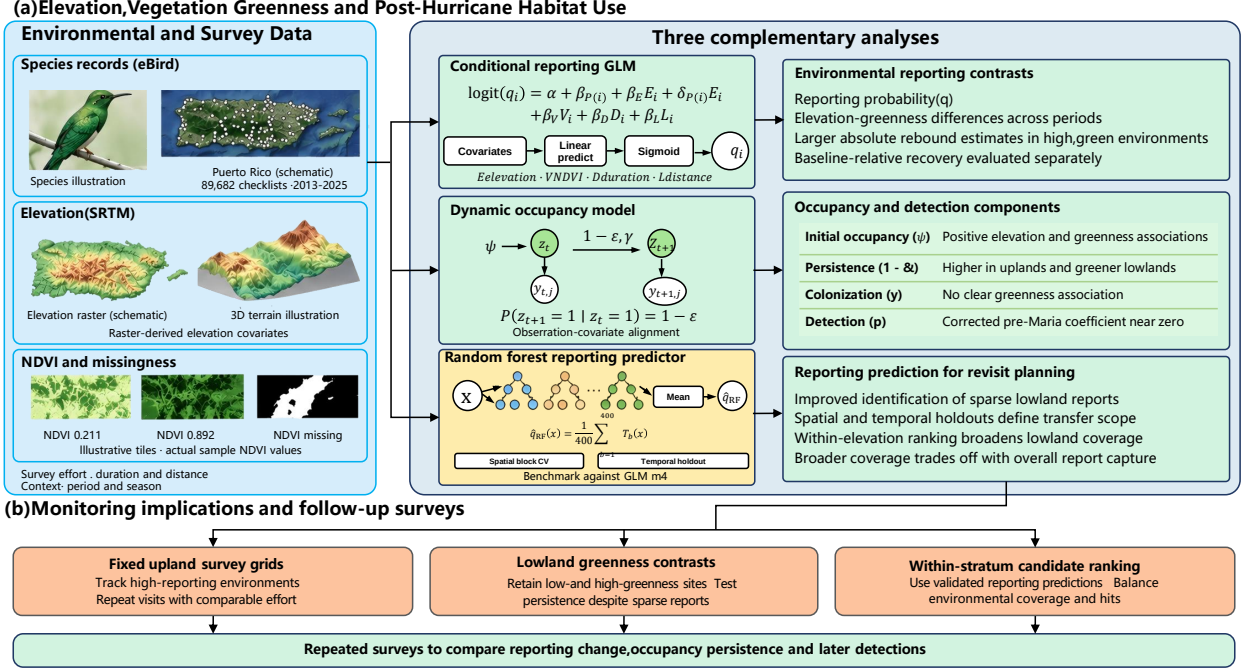


Figure 3: Overview of the environmental analyses and monitoring implications. (a) eBird records, SRTM elevation, NDVI and survey covariates inform complementary analyses of conditional reporting, dynamic occupancy and random forest reporting prediction. (b) The findings motivate fixed upland revisits, lowland greenness contrasts and within-stratum candidate ranking to guide follow-up surveys.

2.6 Neural prediction structure and training-source diagnostics

The archived HERON workflow factorizes a reporting score into two bounded branch scores, $q_i = s_i d_i$, and combines it with conventional models. We reconstruct the archived fold assignments and match sampling-event identifiers between the complete and all-record tables. The original workflow partitions these tables independently but treats equal fold numbers as equal held-out records. We assess overlap with the neural training source before interpreting its performance; internal stacking uses an analogous independent allocation. The diagnostic calculates overlap from the original allocation rules and observation identifiers.

Separately, the grouped loss uses $1 - \prod_i (1 - s_i d_i)$, which depends only on the products. It therefore does not by itself identify either branch as occupancy or detection. For a shared site state, an occupancy model instead has $\Pr(\text{at least one report}) = \psi_g [1 - \prod_j (1 - p_{gj})]$. Input restrictions and monotone effort curves do not establish equivalence to this shared-state likelihood. Accordingly, neural environmental and effort branches are interpreted as fitted scores, while occupancy persistence is evaluated through the dynamic occupancy transitions. Neural-component ablations use 20 folds, stack ablations use six, and the direct configuration comparison uses ten; contrasts are analyzed within their respective fold sets.

3 Results

3.1 Reporting rebounded partially across hurricane-related periods

Of 89,682 checklists, 5,174 reported the species, an overall rate of 5.77%. Reporting declined from 7.59% before María to 4.62% between hurricanes, then increased to 6.21% after Fiona (Table 3; Figure 4). The later increase therefore did not restore the aggregate initial rate. Annual summaries show a low of 3.80% in 2019 and 6.62% in 2025. The period-level rate recovered by 1.59 percentage points but remained 1.38 points below the initial value, indicating a partial rebound following the decline. This aggregate change raises the environmental question of whether rebound is widespread

across previously reporting sites or concentrated in particular elevation and greenness combinations. The stratified comparisons below locate the pattern before site-transition results examine its persistence interpretation.

Table 3: Checklist composition across hurricane-related periods. Reporting rates use all retained checklists; complete records enter the environmental GLM and conventional benchmark.

Period	Checklists	Reports	Rate (%)	NDVI complete	Missing (%)
Pre-Maria	14,728	1,118	7.59	10,689	27.42
Inter-Hurricanes	37,701	1,742	4.62	28,707	23.86
Post-Fiona	37,253	2,314	6.21	29,580	20.60

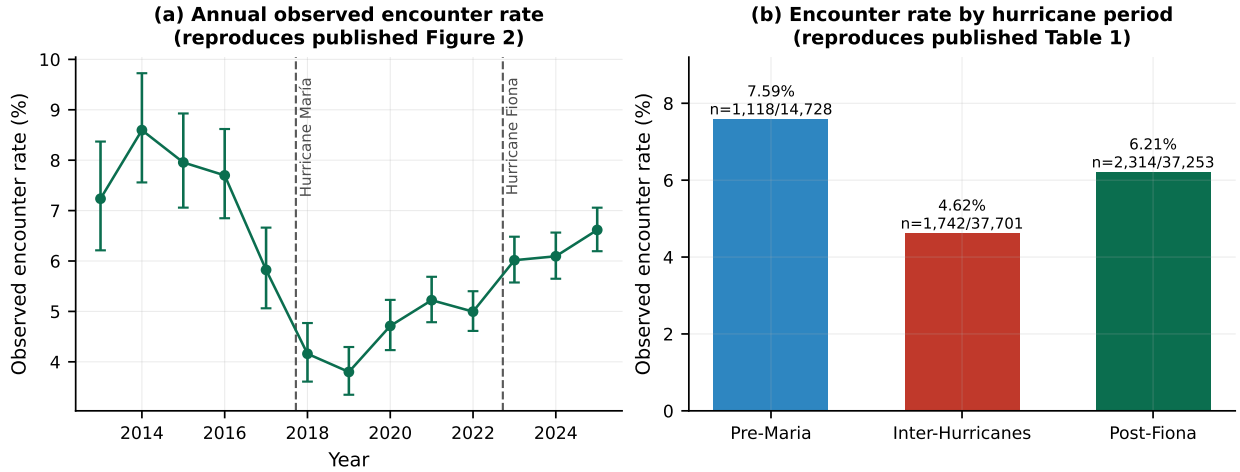


Figure 4: Observed annual and period-level reporting rates from the released records. Fractions printed above the period bars give positive reports divided by all checklists. Annual intervals are the original binomial intervals and do not account for dependence among checklists.

3.2 NDVI missingness concentrates in high-reporting upland environments

NDVI availability increased from 72.58% before María to 79.40% after Fiona. Reporting was 4.89% among complete records (3,370/68,976) and 8.71% among missing records (1,804/20,706). Missingness differed considerably among elevation bands (Table 4; Figure 5). It was 20.8%, 32.8% and 49.6% in low, middle and high bands. High-elevation checklists formed 11.2% of missing records but only 3.4% of complete records. Their complete-record report rate was 37.4%, compared with 3.3% at low elevation. High-reporting upland environments consequently receive greater weight in the missing subset. Conditional comparisons support this compositional explanation. The unadjusted report

Table 4: Elevation-stratified NDVI availability and reporting; rates are percentages.

Elevation	Available n	Missing n	Missing (%)	Report: available (%)	Report: missing (%)
Low	62659	16459	20.8	3.3	4.0
Mid	3970	1937	32.8	11.3	16.6
High	2347	2310	49.6	37.4	35.7

odds ratio for missing records was 1.86 (95% interval 1.51–2.28), declining to 1.03 (0.85–1.26) after elevation and period adjustment. Further adjustment for effort, season and location gave 1.20 (1.03–1.41). Much of the aggregate difference therefore reflects environmental and period composition, with a smaller residual observation difference after fuller adjustment. Within the high-elevation

band, missing records actually had slightly lower reporting than complete records (35.7% versus 37.4%), showing that missingness does not correspond to uniformly greater reporting across environments. For locating rebound, complete records downweight high-reporting uplands while NDVI availability also changes across periods. Raw summaries consequently mix changes in reporting with changes in environmental representation. Stratification and a common covariate distribution improve period comparability; recovering environmental information for missing upland records would help assess whether the higher-persistence association extends to observations omitted from the environmental analysis. For persistence research, complete-case filtering reduces the checklist

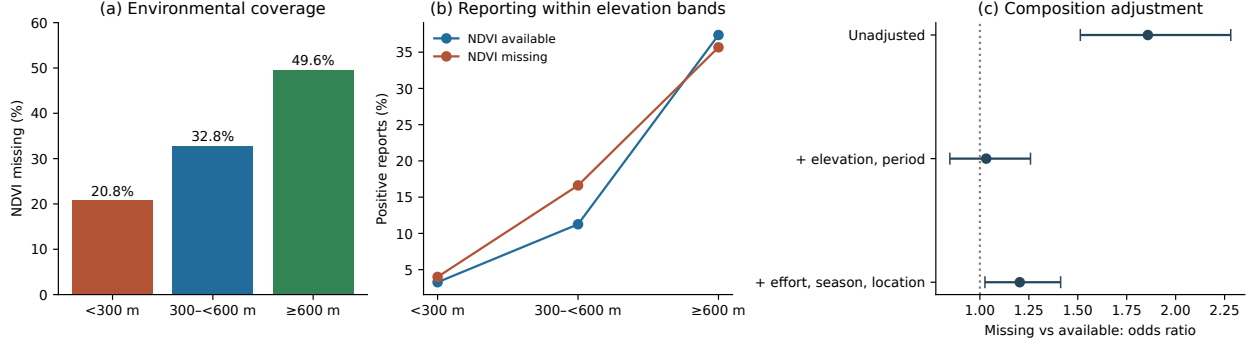


Figure 5: Environmental structure of NDVI missingness. Panels show (a) missing fraction by elevation, (b) reporting within each band, and (c) sequentially adjusted odds ratios with spatial block-clustered 95% intervals.

weight of upland environments rather than removing an environmentally uniform sample. Of 1,048 grids represented in all three periods, only four lacked NDVI entirely; the dynamic model retained 1,044 grids using cross-period mean greenness. Missingness affects the checklist and occupancy analyses differently. Complete-case filtering changes upland record representation in the checklist model, whereas the occupancy model depends on how well cross-period mean vegetation represents conditions in each period. Monitoring should retain repeated upland visits and improve matching to contemporaneous vegetation observations, rather than treating complete-checklist weights as the weights of all sites.

3.3 The period contrast has an elevation structure

The reproduced interaction GLM m4 had AIC 21373.31, model weight 0.776 and pseudo- R^2 0.207 (Table 5). The additive environmental model m3 remained competitive (Δ AIC 2.49), whereas adding environment to the period-and-effort model reduced AIC by 3704.25. Environmental gradients account for substantial fitted variation within this candidate set; the additional evidence for the period-elevation interaction is more limited. Environmental baselines are therefore central to explaining report distributions, whereas locating stronger rebound requires within-environment period contrasts. Persistently high reporting at greater elevation does not itself establish faster recovery.

Table 5: Reproduced candidate GLMs on the same 68,976 NDVI-complete checklists. AIC concerns this fitted candidate set, not geographic predictive performance.

Model	Structure	k	AIC	Δ AIC	w	pseudo- R^2
m4	Period * Elev + NDVI + Dur + Dist	9	21373.31	0.00	0.776	0.2067
m3	Period + Elev + NDVI + Dur + Dist	7	21375.80	2.49	0.224	0.2065
m2	Period + Dur + Dist	5	25077.56	3704.25	0.000	0.0688
m1	Period	3	26796.82	5423.51	0.000	0.0048
m0	1	1	26921.62	5548.31	0.000	0.0000

At fixed covariates, reporting probability at 100 m was 3.77%, 2.40% and 3.08% across the three periods. At 900 m it was 40.58%, 36.52% and 44.89%. Post-Fiona predictions therefore remained below the initial baseline at low elevation while exceeding it at high elevation (Figure 6). These conditional comparisons and the aggregate incomplete rebound describe different quantities and can hold simultaneously. The odds ratios per 100 m were 1.430, 1.483 and 1.500. With the original model-based standard errors, the Post-Fiona interaction had $p = 0.011$; the Inter-Hurricanes interaction had $p = 0.063$.

NDVI was positively associated with reporting ($\beta = 0.871$, SE 0.087). An increase of 0.1 NDVI corresponded to an odds ratio of approximately 1.091 (95% Wald interval 1.073–1.110). The probability-weighted elevation centroid shifted by +28.53 m between hurricanes and +13.88 m after Fiona relative to the initial period. These shifts summarize reweighting of the fitted elevation curve, not individual displacement. The within-Post-Fiona linear time term did not reach the 0.05 level ($p = 0.121$). Period means and annual trends describe different temporal scales; rebound here denotes the increase between period-level reporting rates.

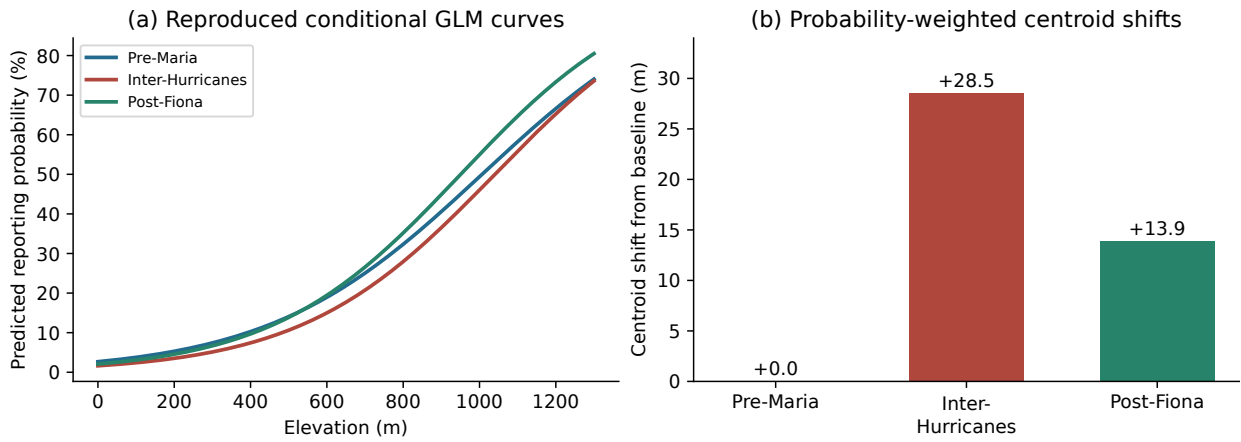


Figure 6: Environmental structure of the reproduced reporting pattern. (a) GLM predictions with NDVI, duration and distance fixed at their complete-sample means. (b) Probability-weighted centroid shifts on the uniform 0–1300 m prediction grid. Curves are fitted reporting responses; centroid shifts do not measure movement.

3.4 Environmental strata locate rebound within different baselines and changes

Standardized predictions across the nine environmental cells show larger environmental contrasts than within-cell period changes (Table 6; Figure 7). Low-elevation, low-greenness probabilities were 2.29%, 1.42% and 1.81% across periods, versus 6.27%, 4.10% and 5.17% at low elevation and high greenness; both partially rebounded after declining. The high-elevation, high-greenness cell gave 42.48%, 38.01% and 44.98%. Higher greenness corresponded to a higher reporting baseline within elevation bands, while upland cells retained high reporting from an already high baseline. From Inter-Hurricanes to Post-Fiona, the low-elevation low- and high-greenness cells increased by 0.39 and 1.07 percentage points, respectively, compared with 6.97 points in the high-elevation, high-greenness cell. By absolute probability change, the larger rebound point estimate therefore occurs in high, green environments. Greener lowlands retain higher reporting levels but remain below their own initial prediction. The subsequent persistence analysis examines whether high upland reporting corresponds to higher occupancy persistence probabilities and whether greener lowlands also have higher persistence despite incomplete aggregate reporting rebound. Spatially clustered intervals show lower precision for the temporal increments. The low-elevation, high-greenness Post-Fiona contrast was -1.10 percentage points (95% interval $[-2.62, 0.42]$), while the high-elevation, high-greenness contrast was $+2.50$ points ($[-5.09, 10.09]$). Coverage was uneven. Only 144 complete checklists represented high elevation with low greenness, versus 22,433 at low elevation with low greenness. The data locate high- and low-reporting environments more precisely than they establish

Table 6: Standardized reporting by elevation/greenness stratum. Periods share the within-stratum covariate distribution; contrast intervals use spatial block-clustered covariance.

Elevation/NDVI	<i>n</i>	Reports	Pre (%)	Inter (%)	Post (%)	Change, pp [95% CI]
Low/Low	22433	423	2.3	1.4	1.8	-0.48 [-1.16, +0.19]
Low/Mid	21009	607	3.8	2.4	3.0	-0.74 [-1.76, +0.27]
Low/High	19217	1016	6.3	4.1	5.2	-1.10 [-2.62, +0.42]
Mid/Low	415	42	10.4	7.7	10.1	-0.33 [-2.64, +1.98]
Mid/Mid	1438	61	13.1	9.9	12.8	-0.34 [-3.18, +2.49]
Mid/High	2117	344	16.8	12.9	16.4	-0.48 [-3.82, +2.87]
High/Low	144	45	23.7	20.3	25.2	+1.46 [-3.73, +6.65]
High/Mid	545	166	30.4	26.5	32.5	+2.05 [-4.44, +8.54]
High/High	1658	666	42.5	38.0	45.0	+2.50 [-5.09, +10.09]

how far each has surpassed its initial level. All cell-specific Post-Fiona versus Pre-María intervals span zero, weakening a categorical claim that uplands have recovered while lowlands have not. Higher baselines also affect the size of absolute increments. Repeated surveys at the same sites with comparable effort should track both absolute and baseline-relative changes and improve coverage of sparse cells, including less-green uplands, to test how widely the fitted gradient extends across sites.

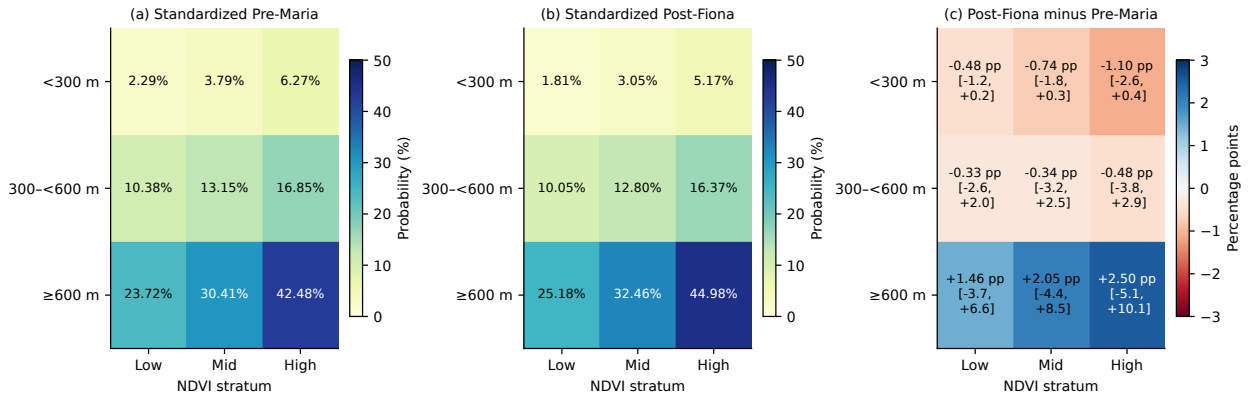


Figure 7: Standardized reporting probabilities and changes by environmental cell. The first two panels show period probabilities; the third gives Post-Fiona minus Pre-María percentage points and spatially clustered 95% intervals. All periods share the same within-cell covariate distribution.

3.5 Persistence associations remain after observation-model correction

The ordering refit converged and changed AIC from 2878.81 to 2892.10. Realignment period labels to the visits at each site changed the fitted observation process. Within the corrected model, initial occupancy remained positively associated with elevation ($\beta = 1.647$) and NDVI ($\beta = 0.583$), while local extinction remained negatively associated with elevation ($\beta = -0.720$) and NDVI ($\beta = -0.818$). Their approximate 95% intervals excluded zero (Table 7). Thus the environmental association with persistence survives this specific implementation correction.

The detection contrast changed more substantially. The Pre-María coefficient relative to Inter-Hurricanes changed from -0.566 (SE 0.190, $p = 0.0029$) to -0.055 (SE 0.159, $p = 0.727$). Corrected detection probabilities were 30.3%, 31.5% and 33.8% across the three periods, with neither period contrast clearly separated from zero (Figure 8). The absolute initial-period coefficient shrank by approximately 90%, and the corrected probabilities spanned 3.5 percentage points. The initial detection contrast was consequently sensitive to label ordering, while environmental persistence associations remained stable. Full published, reproduced and corrected coefficients are retained in Table 8.

Table 7: Corrected dynamic occupancy coefficients. Environmental predictors are standardized across retained sites. Detection contrasts use Inter-Hurricanes as reference. Intervals are approximate 95% Wald intervals.

Parameter	Estimate	SE	95% interval	<i>p</i>
Initial occupancy: elevation	1.647	0.268	[1.122, 2.171]	< 0.001
Initial occupancy: NDVI	0.583	0.179	[0.231, 0.934]	0.001
Colonization: NDVI	0.057	0.171	[-0.278, 0.392]	0.740
Local extinction: elevation	-0.720	0.238	[-1.187, -0.252]	0.003
Local extinction: NDVI	-0.818	0.313	[-1.431, -0.206]	0.009
Detection: Pre-Maria	-0.055	0.159	[-0.366, 0.256]	0.727
Detection: Post-Fiona	0.107	0.149	[-0.185, 0.398]	0.474

Table 8: Published, reproduced and ordering-corrected dynamic occupancy estimates. The controlled refit changes only observation-covariate alignment (Section 2.3). Corrected uncertainty is summarized in Table 7.

Comp.	Parameter	Published		Authors' script (ours)		Corrected order	
		est.	SE	est.	SE	est.	<i>p</i>
ψ	(Intercept)	-1.637	0.171	-1.6365	0.1710	-1.6287	1.06e-17
ψ	Elevation	1.544	0.238	1.5436	0.2379	1.6466	7.59e-10
ψ	NDVI	0.613	0.177	0.6127	0.1766	0.5825	0.00115
γ	(Intercept)	-2.701	0.172	-2.7014	0.1723	-2.8155	4.96e-56
γ	NDVI	0.074	0.170	0.0737	0.1704	0.0566	0.74
ε	(Intercept)	-0.177	0.306	-0.1768	0.3062	-0.0905	0.763
ε	Elevation	-0.830	0.296	-0.8299	0.2963	-0.7197	0.00254
ε	NDVI	-0.840	0.337	-0.8400	0.3372	-0.8185	0.00884
p	(Intercept)	-0.736	0.119	-0.7364	0.1191	-0.7780	7.57e-12
p	PeriodPost-Fiona	0.120	0.146	0.1200	0.1460	0.1066	0.474
p	PeriodPre-Maria	-0.566	0.190	-0.5661	0.1904	-0.0553	0.727
AIC		2878.81		2878.8085		2892.1042	

The NDVI colonization coefficient was close to zero ($\beta = 0.057$, 95% interval $[-0.278, 0.392]$). Evidence linking greenness to occupancy persistence is therefore clearer than evidence linking it to entry into unoccupied sites. This strengthens the interpretation linking high reporting in greener environments to retention of occupied sites, while providing less support for attributing rebound to greenness-associated colonization. Within this model, greenness primarily characterized retention of occupied sites, whereas a single greenness gradient explained entry into occupancy weakly. Monitoring should retain detections and nondetections from each visit to estimate occupancy persistence and colonization probabilities separately.

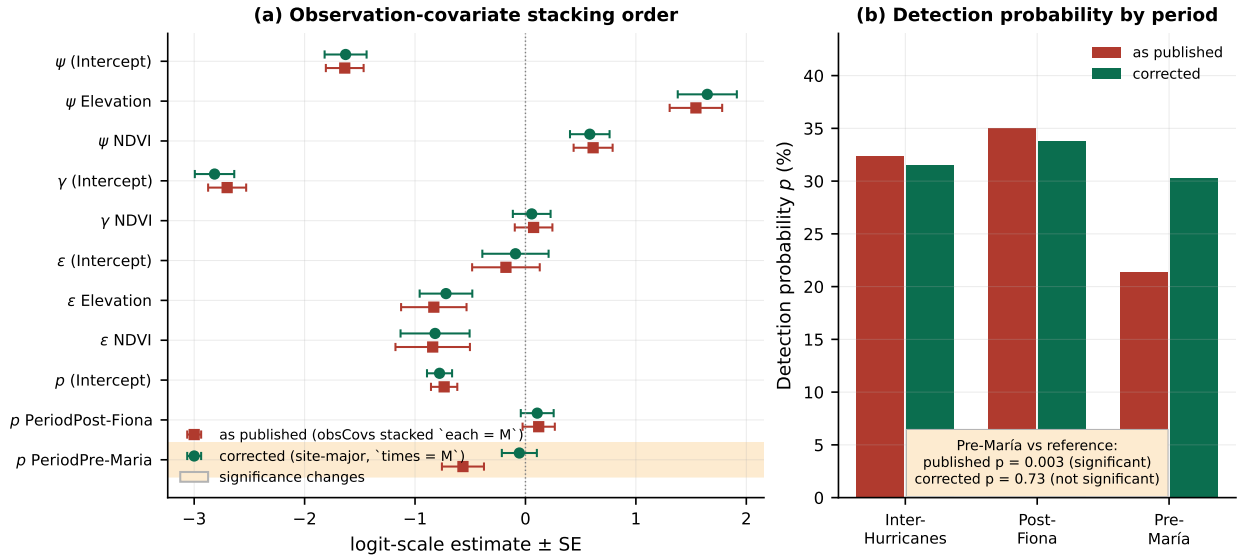


Figure 8: Effect of correcting period-label alignment in the selected dynamic occupancy model. The archived coefficient plot uses one-SE error bars; Table 7 gives 95% intervals. Detection probabilities follow the panel’s displayed period order. Environmental and detection coefficients differ in their sensitivity to observation-order correction.

3.6 Low-elevation occupancy persistence varies with greenness

Projecting corrected transition probabilities onto observed site environments reveals a persistence structure beyond the coefficient signs (Table 9; Figure 9). Mean occupancy persistence probability was 24.2%, 49.7% and 71.3% across low-elevation low-, middle- and high-greenness cells, containing 320, 430 and 158 sites. The 39 high-elevation, high-greenness sites averaged 96.8%. Persistence therefore varies both between uplands and lowlands and along greenness within lowlands. Compared with the reporting strata, high-elevation, high-greenness environments combine high reporting with high modeled persistence, identifying environments in which fixed survey grids can test occupancy in later periods. In greener lowlands, the reporting point estimate remains below its initial baseline despite relatively high occupancy persistence probability. Rebound and retention of occupied sites describe different processes, so low aggregate reporting does not imply uniformly weak persistence across lowlands. These probabilities express environmental associations derived from the same corrected model. Holding elevation at 100 m, increasing NDVI from 0.30 to 0.76 raised predicted

Table 9: Occupancy persistence, colonization, and repeated reports by environment. Occupancy persistence is conditional on the grid being occupied in the previous period; intervals use 2,000 joint parameter draws.

Elevation/NDVI	Grids	Persistence % [95% CI]	Colonization % [95% CI]	Repeated reports
Low/Low	320	24.2 [11.0, 46.4]	5.3 [3.6, 7.9]	1/9
Low/Mid	430	49.7 [35.8, 63.2]	5.7 [4.1, 8.1]	14/46
Low/High	158	71.3 [53.4, 83.9]	6.0 [3.4, 10.4]	15/28
Mid/Low†	5	61.8 [33.4, 83.6]	5.4 [3.9, 7.5]	0/1
Mid/Mid	24	79.9 [65.0, 89.2]	5.8 [4.0, 8.4]	3/4
Mid/High	48	89.3 [79.3, 94.6]	6.1 [3.3, 10.9]	17/27
High/Low†	2	77.8 [31.5, 95.3]	5.3 [3.5, 7.9]	0/0
High/Mid	18	94.2 [79.6, 98.4]	5.8 [3.9, 8.7]	11/14
High/High	39	96.8 [88.7, 99.0]	6.1 [3.3, 10.9]	22/32

† Fewer than ten grids. Repeated reports use grid–period pairs with three visits in each adjacent period; the denominator counts pairs with a report in the previous period.

occupancy persistence probability from 29.8% to 72.0%, a 42.2-point contrast (95% interval 9.9–64.9). Colonization changed from 5.3% to 6.0%, a 0.7-point contrast with interval [−2.9, 5.3].

Holding elevation fixed supports an interpretation in which greenness chiefly characterizes retention of occupied sites. Maintaining vegetation conditions at occupied sites and promoting transitions from nonoccupancy to occupancy are therefore distinct monitoring targets; the retention association does not directly quantify a colonization benefit from increasing vegetation. Equal-visit histories also

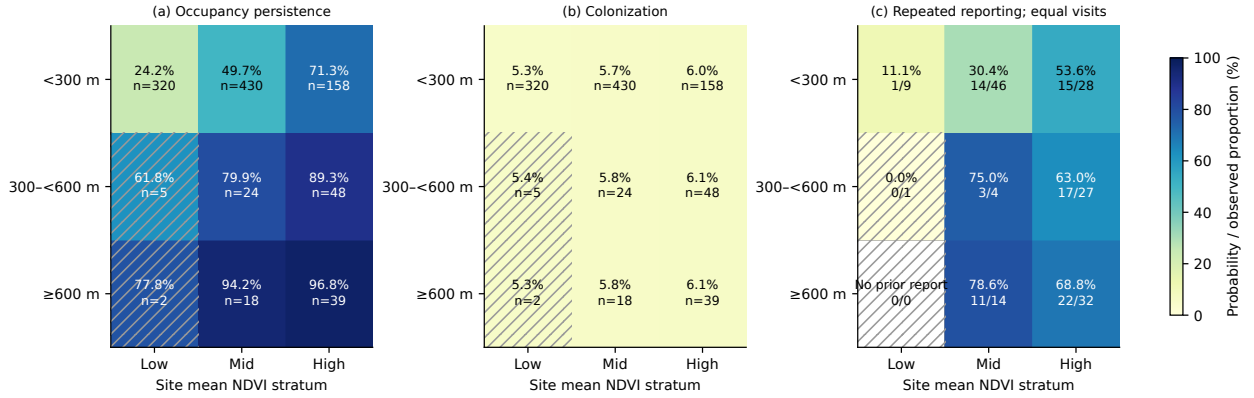


Figure 9: Corrected-model processes and equal-visit reporting histories. Panels show (a) occupancy persistence probability, (b) colonization, and (c) repeated reports conditional on a previous report, with counts. Site counts appear in the first two panels; hatching denotes fewer than ten sites. Process probabilities concern transitions between model primary periods.

differed across environments. Repeated reports occurred in 1/9 previously reporting low-elevation, low-greenness period pairs, 15/28 low-elevation, high-greenness pairs, and 22/32 high-elevation, high-greenness pairs. Higher repeat-report proportions in greener environments are directionally consistent with higher modeled occupancy persistence; repeat reports depend on both presence and detection. Among ten fully sampled high-elevation, high-greenness pairs with no earlier report, seven had a later report, despite modeled colonization near 6%. Frequent later reports among previously nonreporting pairs alongside low modeled colonization illustrates that nonreporting sites can still be occupied under the detection model. Repeated surveys are therefore valuable for distinguishing renewed detection at occupied grids from transitions out of nonoccupancy.

3.7 Reporting prediction retains value under geographic holdout

All eleven conventional configurations were evaluated on the same 30 spatial test folds (Table 10). The published GLM m4 had mean AUC 0.800, AP 0.253 and Brier score 0.0401. Terrain-augmented LightGBM had the highest mean AUC, 0.846. Random forest had the highest mean AP, 0.286, and lowest mean Brier score, 0.0384. Its top-ranked 10% of test checklists contained 53.4% of positive reports, compared with 46.1% for m4. Relative to m4, random forest increased AP by approximately 13.1%, reduced Brier by approximately 4.3% and increased top-decile recall by 7.3 percentage points. At the same checklist-screening fraction, more positive reports entered the priority set, demonstrating its value for ranking monitoring records. Model rankings can identify reported locations worth revisiting, after which repeated surveys and detection models can compare occupancy persistence probabilities across elevation and greenness conditions. The following stratified analysis examines which environments benefit from improved report identification.

The changes were selective (Figure 10). Adding terrain to m4 increased AUC from 0.800 to 0.822, while AP changed only from 0.253 to 0.255 and Brier error increased slightly. Terrain improved the boosting configurations but reduced the MLP’s performance. The additive m3 and interaction m4 GLMs had almost identical spatial results despite the latter’s better sample-fit AIC. Thus an environmental term can help describe the sampled pattern without providing a corresponding improvement in held-out report prediction. Different feature sets also prevent attributing tree-model gains solely to model architecture. For monitoring, terrain inputs should be judged by whether they improve positive-report coverage in target environments. Higher AUC with nearly unchanged AP indicates that stronger overall discrimination need not supply more useful revisit clues under

Table 10: Conventional models on the same 30 spatial test folds. Entries are unweighted means $\pm$ between-fold SD; SD is not a confidence interval. Models use different feature sets.

Model	AUC	AP	Brier	Top-10% recall
GLM-m4	0.800 $\pm$ 0.071	0.253 $\pm$ 0.124	0.0401 $\pm$ 0.0183	0.461 $\pm$ 0.124
GLM-m3	0.801 $\pm$ 0.071	0.254 $\pm$ 0.124	0.0401 $\pm$ 0.0183	0.462 $\pm$ 0.125
GLM-splines	0.812 $\pm$ 0.070	0.260 $\pm$ 0.120	0.0397 $\pm$ 0.0176	0.493 $\pm$ 0.140
GLM-m4 + terrain	0.822 $\pm$ 0.065	0.255 $\pm$ 0.114	0.0403 $\pm$ 0.0190	0.489 $\pm$ 0.130
Random forest	0.842 $\pm$ 0.055	0.286 $\pm$ 0.120	0.0384 $\pm$ 0.0173	0.534 $\pm$ 0.129
LightGBM	0.839 $\pm$ 0.051	0.262 $\pm$ 0.117	0.0397 $\pm$ 0.0185	0.506 $\pm$ 0.140
LightGBM + terrain	0.846 $\pm$ 0.050	0.270 $\pm$ 0.111	0.0396 $\pm$ 0.0189	0.513 $\pm$ 0.133
XGBoost	0.840 $\pm$ 0.051	0.269 $\pm$ 0.119	0.0392 $\pm$ 0.0181	0.517 $\pm$ 0.137
XGBoost + terrain	0.844 $\pm$ 0.051	0.273 $\pm$ 0.115	0.0389 $\pm$ 0.0182	0.511 $\pm$ 0.130
MLP	0.777 $\pm$ 0.065	0.188 $\pm$ 0.102	0.0494 $\pm$ 0.0222	0.418 $\pm$ 0.130
MLP + terrain	0.748 $\pm$ 0.068	0.168 $\pm$ 0.094	0.0564 $\pm$ 0.0297	0.386 $\pm$ 0.107

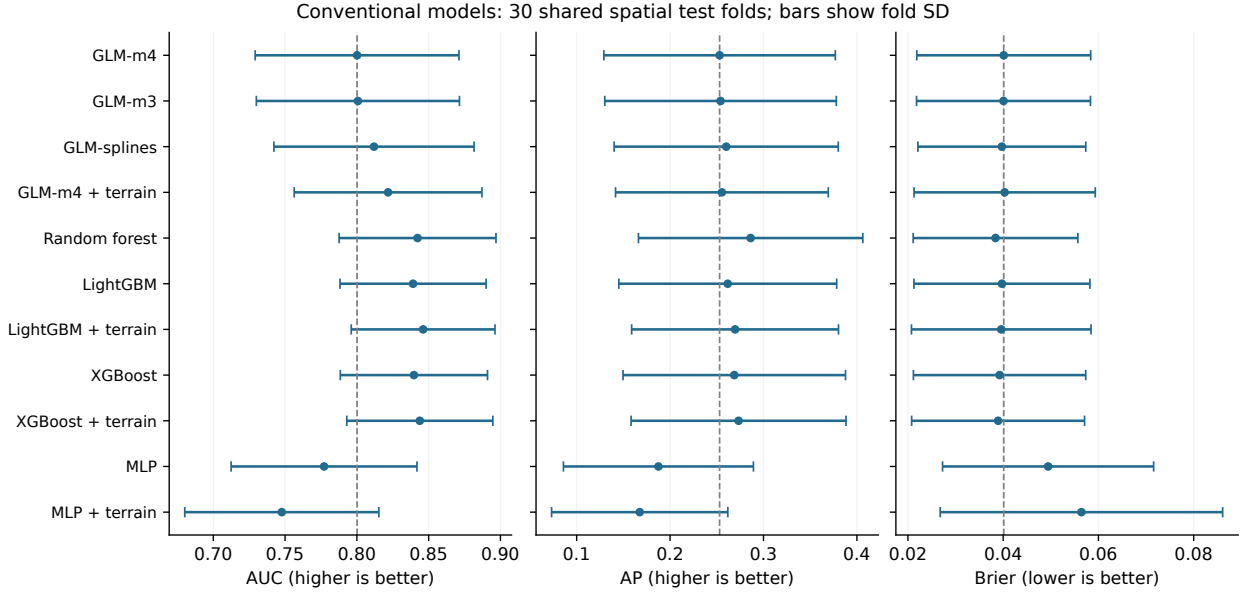


Figure 10: Conventional reporting models on 30 shared geographic test folds. Points are fold means and bars are between-fold SD, not confidence intervals. Dashed lines mark m4. The best mean overall discrimination, positive-report ranking and probability error are not all supplied by the same configuration.

limited screening capacity. Similar m3 and m4 performance also shows that an interaction describing rebound gradients need not improve site prioritization.

3.8 Better identification of lowland reports guides revisits for occupancy monitoring

Lowlands are important for testing the connection between reporting rebound and occupancy persistence. Their reporting rate was only 3.3%, yet they contained 60.7% of complete-sample positive reports, while modeled persistence was 24.2% and 71.3% in low- and high-greenness lowland cells. Low aggregate reporting therefore conceals differences in occupancy persistence probabilities within lowlands. In the new ten-fold experiment, random forest increased low-elevation AP from m4's 0.143 to 0.201 and middle-elevation AP from 0.221 to 0.326; high-elevation AP changed from 0.540 to 0.538 (Table 11; Figure 11). Prediction gains chiefly improve identification of sparse low- and middle-elevation reports, providing a basis for selecting candidate revisit locations from existing records. Higher modeled persistence concentrates in uplands and greener sites, whereas larger

predictive gains concentrate at low and middle elevations; these patterns supply complementary monitoring information.

Table 11: Environment-stratified prediction and monitoring coverage. Both policies select 6,890 checklists: Global ranks the top 10% within each fold, and Stratified allocates approximately 10% within each elevation band and fold.

Elevation	Model	Reports	AP	Brier	Global selected %	Global recall %	Stratified recall %
Low	GLM m4	2046	0.143	0.0299	6.1	26.4	38.2
Low	RF	2046	0.201	0.0289	6.9	34.6	45.6
Mid	GLM m4	447	0.221	0.0974	38.0	53.5	19.5
Mid	RF	447	0.326	0.0868	29.7	52.3	26.2
High	GLM m4	877	0.540	0.2168	66.6	78.3	16.4
High	RF	877	0.538	0.2154	57.9	72.3	15.7

Under global top-decile ranking, random forest captured 34.6% of low-elevation positives versus 26.4% for m4, an 8.1-point gain with spatial block bootstrap interval 2.8–12.0. Identifying additional lowland reports expands the candidate pool for repeated surveys that can compare repeat-report proportions across greenness conditions and estimate occupancy persistence after accounting for imperfect detection. Report identification and persistence estimation require separate steps. Checklist ranking does not measure occupancy after a revisit. High-elevation positive coverage simultaneously declined from 78.3% to 72.3%, a -6.0 -point contrast with interval $[-17.2, -0.9]$. Changing models therefore improved lowland identification while reducing upland report coverage. Retaining fixed upland repeat sites would prevent this ranking shift from weakening longitudinal coverage of environments with high modeled persistence.

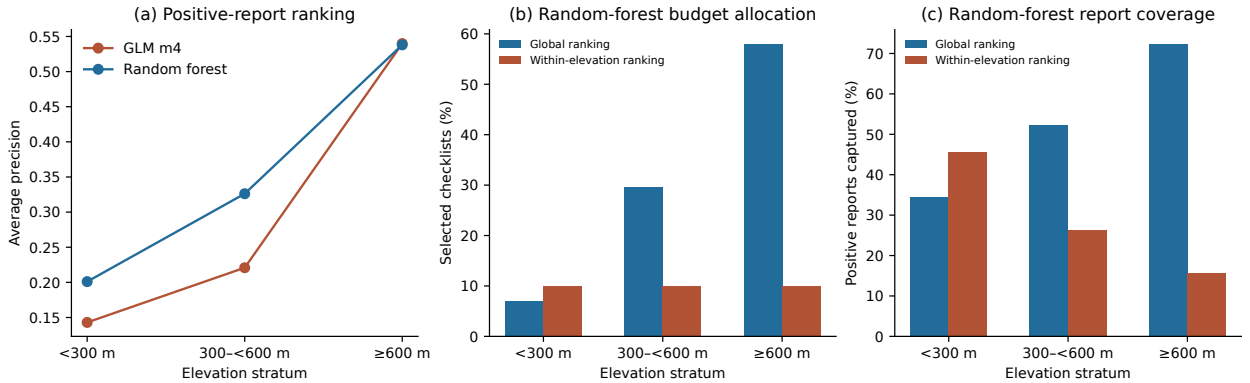


Figure 11: Prediction gains and monitoring coverage. Panels show (a) AP by elevation, (b) within-band selection under two equal-budget random-forest policies, and (c) within-band positive-report coverage. Each policy selects 6,890 checklists.

The equal-budget comparison shows how the monitoring objective changes environmental coverage. Global random-forest ranking selected 6.9%, 29.7% and 57.9% of low-, middle- and high-elevation checklists, concentrating screening slots in high-reporting environments. Selecting approximately 10% within each elevation band increased lowland positive coverage from 34.6% to 45.6%, an 11.0-point gain (95% interval 7.4–15.5), while reducing overall coverage from 46.7% to 35.2%, or 388 fewer positive reports. Wider lowland representation supplies a broader record base for comparing incomplete reporting rebound with occupancy persistence, but reduces aggregate hits under the same screening budget. Environmental monitoring can combine fixed upland revisits with lowland coverage and use within-stratum ranking to identify candidate sites, retaining contrasting greenness conditions within lowlands. The experiment quantifies checklist coverage under elevation stratification; greenness allocation and actual field costs require a separate survey design. These pooled rates come from a single ten-fold experiment and are reported separately from the earlier unweighted 30-fold averages.

3.9 Spatial validation defines the geographic scope of revisit guidance

Random validation yielded larger average performance values for the illustrated conventional configurations (Figure 12). Random forest’s AUC/AP were 0.910/0.442 under random splitting and 0.842/0.286 under spatial blocking; m4 changed from 0.820/0.299 to 0.800/0.253. The flexible model’s advantage was therefore smaller in the geographic holdout. Random forest’s AP advantage narrowed from 0.143 under random splitting to 0.033 under spatial splitting. This contraction weakens the basis for exporting high random-validation scores to new locations. Together with the preceding lowland analysis, the remaining spatial-holdout gains support using models to identify revisit candidates within the observed environmental domain. However, aggregate spatial performance does not establish equal gains in every lowland greenness cell. Site selection should preserve elevation and greenness coverage before applying a ranking tool assessed under geographic holdout, so that easily predicted locations do not replace the environmental contrasts needed to estimate occupancy persistence. Training fractions and sample composition also differ between protocols; the entire score decline cannot be attributed to geographic distance.

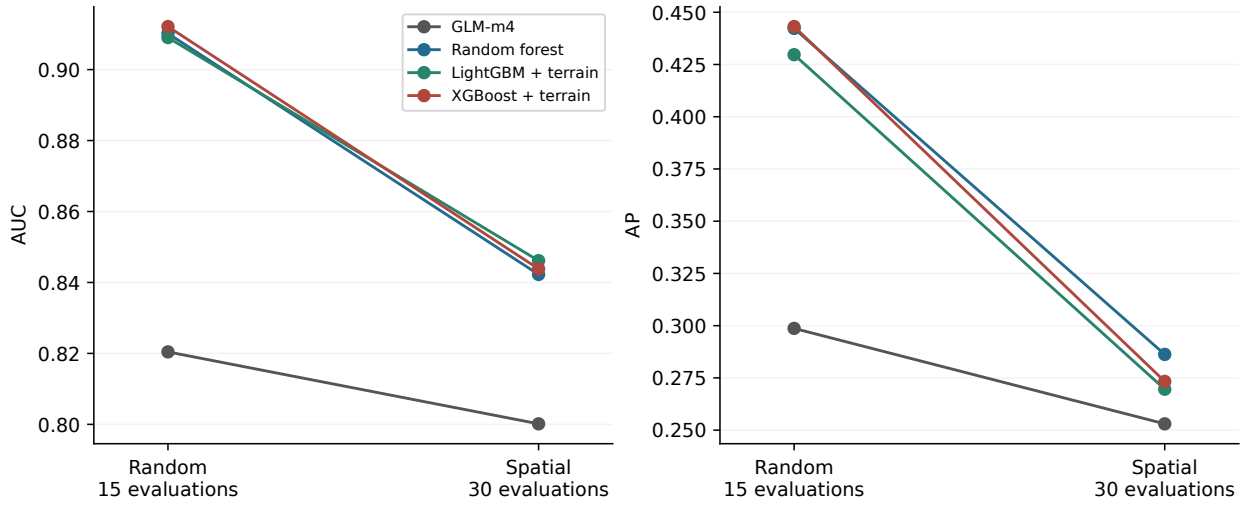


Figure 12: Conventional-model means under random and spatial validation. Random validation has 15 evaluations and spatial validation 30. Lines connect protocol summaries, not paired treatment effects. The protocols differ in training fraction as well as geographic composition.

3.10 Temporal holdout supports identification of later-period revisit candidates

The random, spatial and temporal protocols expose different prediction demands (Figure 13; Table 12). Random splits retain similar location mixtures in training and testing, allowing tree models to use detailed local patterns. Their smaller spatial-holdout advantage indicates weaker transfer of those patterns across locations. In the Post-Fiona time test, random forest achieved AUC 0.8901, AP 0.3869 and Brier 0.0391, compared with 0.8321, 0.3174 and 0.0422 for m4 (Table 13). Top-decile recall increased from 53.28% to 59.80% while probability error declined. Environmental and observation relationships learned in earlier periods therefore retained predictive value for later reports. Rankings learned from earlier records still concentrate positive reports in Post-Fiona observations and can help select sites worth surveying again. In conjunction with the lowland analysis, its monitoring role is to broaden searches for sparse reports before repeated visits supply detection histories for estimating occupancy states and transitions. The temporal test pools elevation bands and does not separately quantify transfer within lowland greenness cells; fixed revisits across those cells are needed to assess whether the predictive information applies equally. Because the test uses the same island observation network, it supports cross-period use of reporting clues, while inference about continued occupancy rests on repeated surveys and transition models.

Table 12: Predictive performance under random validation: ten folds for HERON-plus and 15 for the other configurations.

Method	AUC ↑	AP ↑	pseudo- R^2 ↑	Tjur R^2 ↑	Brier ↓	LogLoss ↓	ECE ↓	Recall at 10pct ↑
GLM-m3 (published)	0.8204	0.2989	0.2059	0.1521	0.0397	0.1550	0.0055	0.5072
GLM-m4 (published) ‡	0.8204	0.2987	0.2060	0.1518	0.0397	0.1549	0.0058	0.5076
GLM-splines	0.8365	0.3237	0.2348	0.1652	0.0385	0.1493	0.0063	0.5515
GLM-m4 + terrain	0.8503	0.3258	0.2440	0.1789	0.0386	0.1475	0.0054	0.5554
RandomForest	0.9102	0.4423	0.3535	0.2458	0.0344	0.1262	0.0055	0.6671
MLP	0.8540	0.3084	0.0452	0.2648	0.0432	0.1863	0.0280	0.5500
MLP + terrain	0.8511	0.3065	-0.4301	0.3003	0.0477	0.2791	0.0402	0.5629
LightGBM	0.9066	0.4250	0.3225	0.2709	0.0355	0.1322	0.0113	0.6487
LightGBM + terrain	0.9090	0.4297	0.3117	0.2813	0.0357	0.1343	0.0142	0.6504
XGBoost	0.9088	0.4311	0.3492	0.2658	0.0348	0.1270	0.0050	0.6561
XGBoost + terrain	0.9121	0.4431	0.3544	0.2799	0.0346	0.1260	0.0065	0.6674
HERON-NN	0.8328	0.3192	0.2294	0.1678	0.0386	0.1504	0.0066	0.5474
HERON	0.9106	0.4308	0.3477	0.2397	0.0348	0.1273	0.0072	0.6568
HERON-plus	0.9362	0.4882	0.3900	0.2905	0.0330	0.1190	0.0149	0.7135

‡ the model published by [5].

Table 13: Post-Fiona temporal holdout, averaged across three fitting seeds evaluated on the same temporal test set.

Method	AUC ↑	AP ↑	pseudo- R^2 ↑	Tjur R^2 ↑	Brier ↓	LogLoss ↓	ECE ↓	Recall at 10pct ↑
GLM-m3 (published)	0.8330	0.3171	0.2203	0.1688	0.0423	0.1622	0.0077	0.5335
GLM-m4 (published) ‡	0.8321	0.3174	0.2189	0.1578	0.0422	0.1624	0.0093	0.5328
GLM-splines	0.8467	0.3389	0.2414	0.1653	0.0411	0.1578	0.0141	0.5711
GLM-m4 + terrain	0.8513	0.3351	0.2462	0.1922	0.0413	0.1568	0.0095	0.5609
RandomForest	0.8901	0.3869	0.3084	0.2096	0.0391	0.1438	0.0063	0.5980
MLP	0.8259	0.2305	-0.1468	0.2017	0.0523	0.2385	0.0400	0.4474
MLP + terrain	0.8328	0.2481	-0.6514	0.2637	0.0583	0.3434	0.0517	0.4929
LightGBM	0.8797	0.3514	0.1965	0.2170	0.0420	0.1671	0.0219	0.5626
LightGBM + terrain	0.8842	0.3545	0.1749	0.2183	0.0421	0.1716	0.0224	0.5670
XGBoost	0.8835	0.3647	0.2730	0.2263	0.0405	0.1512	0.0133	0.5747
XGBoost + terrain	0.8873	0.3712	0.2751	0.2293	0.0402	0.1508	0.0132	0.5847
HERON-NN	0.8391	0.3251	0.2281	0.1871	0.0417	0.1605	0.0147	0.5528
HERON	0.8913	0.3851	0.3109	0.2144	0.0392	0.1433	0.0064	0.5919
HERON-plus	0.8932	0.3949	0.3125	0.2328	0.0390	0.1430	0.0108	0.6029

‡ the model published by [5].

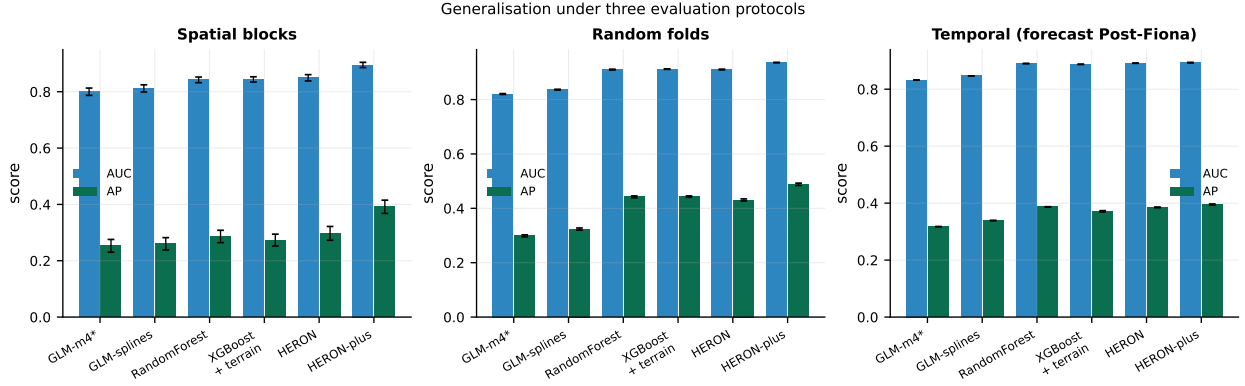


Figure 13: AUC and AP under geographic holdout, observation interpolation and transfer across periods. Neural training-source overlap is analyzed in Section 3.13.

3.11 Block size and vegetation index affect the size of predictive gains

As block width increased from 0.04° to 0.32° , terrain-augmented LightGBM declined from AUC 0.8555 to 0.8351 and from AP 0.3066 to 0.2592 (Figure 14). The corresponding m4 AUC values were 0.8002 and 0.8024. LightGBM’s AUC advantage consequently narrowed from 0.0553 to 0.0327, indicating that part of its predictive information depends on finer spatial organization. At 0.16° , LightGBM retained higher AUC than m4 (0.8413 versus 0.8146) but lower AP (0.2904 versus 0.2975). Better overall discrimination thus did not consistently concentrate rare positive reports at the top of the ranking. Nonmonotonic changes reflect the joint effects of block aggregation and fold composition, making the monitoring scale relevant to model assessment. For lowland persistence surveys, this supports assessing candidate-site rankings at the intended regional scale. Terrain and location information useful among nearby sites may weaken across larger blocks. Fixed sites in each environmental stratum can preserve revisit coverage as the survey domain expands; these pooled scale tests do not separately establish transfer distances for lowland greenness cells. The six-fold

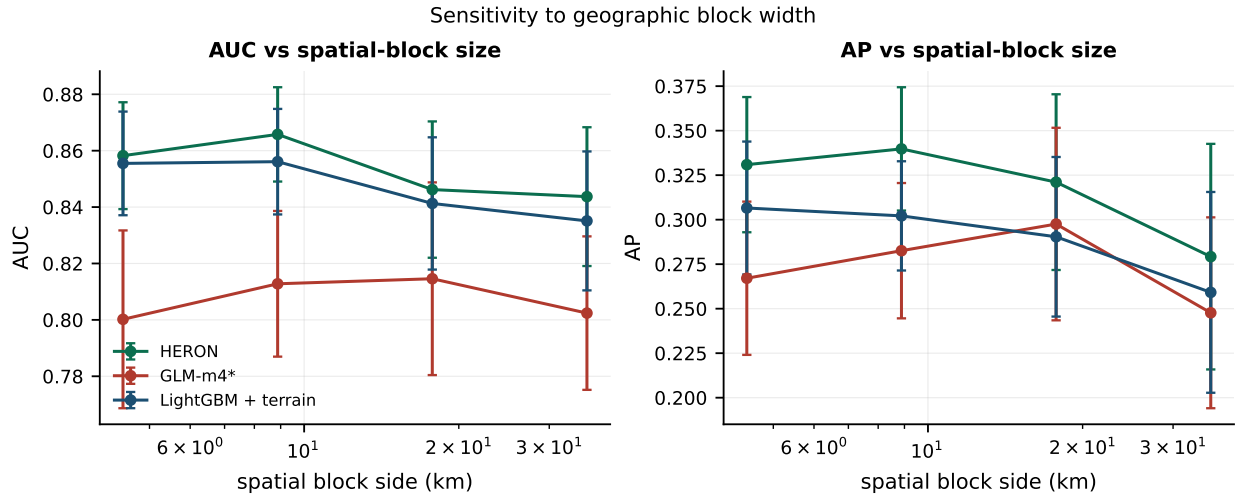


Figure 14: AUC and AP across four geographic block widths, with six folds at each width. Bars show standard errors across folds; no distance buffer was imposed.

EVI2 experiment retained the tree models’ ranking advantage (Table 14). Random forest achieved AP 0.3223, exceeding m4 within the same experiment (0.2765) by 0.0458, while reducing Brier from 0.0402 to 0.0386. Terrain-augmented XGBoost had the highest conventional-model AUC (0.8576) and top-decile recall of 56.15%. The tree-model advantage therefore also appeared with EVI2. In contrast, MLP Brier increased from 0.0515 to 0.0583 when terrain was added. Across the vegetation-index experiments, performance depended on how models used environmental information, rather

than simply on the number of inputs. Retaining useful report ranking with an alternative vegetation index strengthens the basis for vegetation-informed screening of candidate locations. However, this experiment did not re-estimate the association between EVI2 and persistence. The value of greenness for screening and its association with occupancy persistence therefore rest on prediction and transition evidence, respectively.

Table 14: Six-fold spatial prediction means in the EVI2 substitution experiment.

Method	AUC ↑	AP ↑	pseudo- R^2 ↑	Tjur R^2 ↑	Brier ↓	LogLoss ↓	ECE ↓	Recall at 10pct ↑
GLM-m3 (published)	0.8155	0.2770	0.1970	0.1444	0.0402	0.1574	0.0129	0.4713
GLM-m4 (published) ‡	0.8152	0.2765	0.1963	0.1440	0.0402	0.1575	0.0132	0.4716
GLM-splines	0.8280	0.2831	0.2098	0.1478	0.0398	0.1549	0.0169	0.5191
GLM-m4 + terrain	0.8366	0.2913	0.2108	0.1714	0.0406	0.1554	0.0174	0.5174
RandomForest	0.8508	0.3223	0.2499	0.1621	0.0386	0.1476	0.0151	0.5582
MLP	0.7790	0.1953	-0.1964	0.1836	0.0515	0.2369	0.0428	0.4090
MLP + terrain	0.7663	0.1955	-1.0102	0.2049	0.0583	0.3999	0.0544	0.4149
LightGBM	0.8462	0.2825	0.2043	0.1612	0.0401	0.1576	0.0152	0.5341
LightGBM + terrain	0.8559	0.3117	0.2014	0.1675	0.0393	0.1589	0.0186	0.5582
XGBoost	0.8506	0.2984	0.2420	0.1727	0.0393	0.1498	0.0101	0.5554
XGBoost + terrain	0.8576	0.3166	0.2475	0.1845	0.0390	0.1494	0.0144	0.5615
HERON-NN	0.8286	0.2961	0.2218	0.1421	0.0393	0.1530	0.0152	0.5328
HERON	0.8650	0.3387	0.2663	0.1835	0.0381	0.1448	0.0150	0.5832

‡ the model published by [5].

3.12 Ranking and calibration favor different model properties

The pooled ten-fold ROC and precision–recall curves clarify the ranking differences (Figure 15). Pooled AUC/AP were 0.8088/0.2836 for m4, 0.8640/0.3162 for terrain-augmented XGBoost and 0.8625/0.3086 for terrain-augmented LightGBM. Both boosting models improved ordering between positive and nonreporting checklists; XGBoost provided higher AP in the positive-report ranking. With reports comprising only 4.89% of complete records, precision–recall curves directly describe the concentration of positives among highly ranked checklists. Calibration produced a different

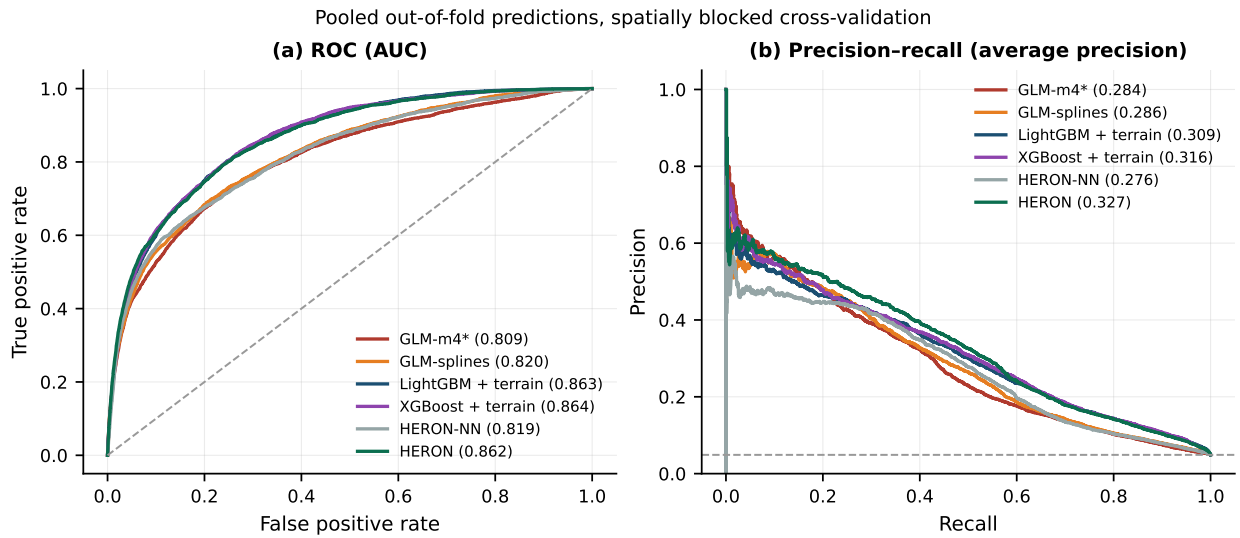


Figure 15: ROC and precision–recall curves pooled across the ten-fold prediction export. Parentheses give pooled AUC or AP; the horizontal dashed line marks report prevalence.

ordering (Figure 16). Expected calibration error was 0.0032 for m4, versus 0.0050 for the spline

GLM, 0.0093 for terrain-augmented XGBoost and 0.0181 for terrain-augmented LightGBM. XGBoost nevertheless had lower pooled Brier than m4 (0.0389 versus 0.0402). Its stronger discrimination reduced overall squared error while grouped predicted probabilities deviated more from observed frequencies. AP and top-decile recall consequently address prioritization, whereas probability calibration also matters when aggregating expected report counts. Ranking helps identify sparse lowland reports, while calibration affects estimated reporting levels within environmental strata. Upward-biased probabilities could overstate apparent rebound if interpreted directly as change. The current pooled calibration results require within-stratum assessment before comparing expected report counts between lowlands and uplands.

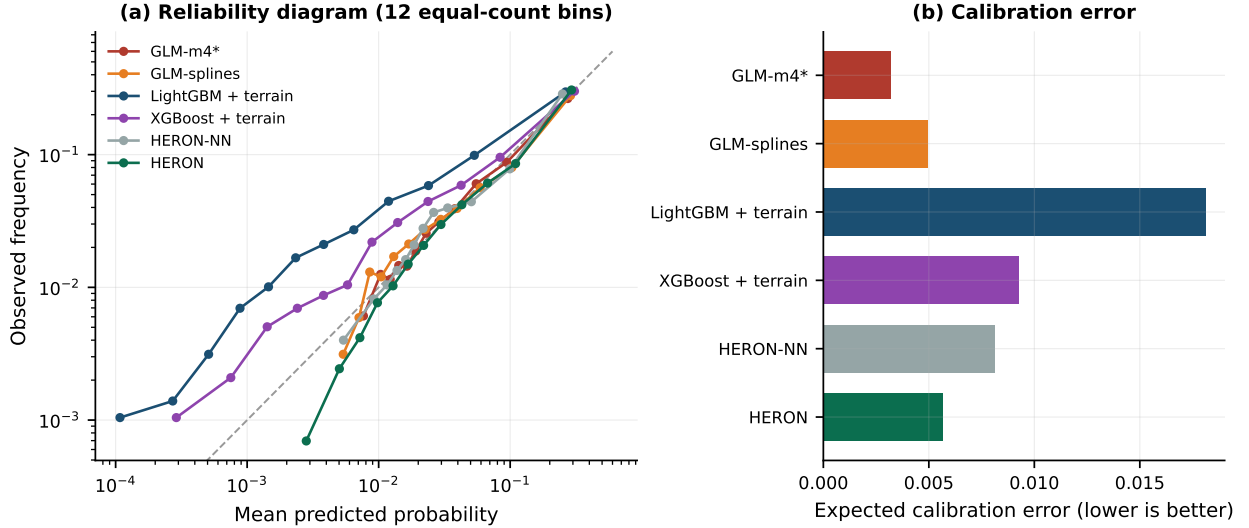


Figure 16: Probability reliability and calibration error. Reliability curves use 12 equal-count bins and expected calibration error uses 15; the diagonal denotes agreement between predicted probability and observed frequency.

3.13 Training-source overlap compromises neural performance estimates

Across three reconstructed spatial partitions, 78.18%, 67.13% and 70.00% of complete-record test observations appeared in the corresponding all-record neural training sources (Figure 17). Independently balancing folds in the two tables assigned different fold identifiers to the same geographic blocks. Subsequent filtering by identifier admitted test records to training. HERON-plus, which omitted internal early-stopping holdout, fitted these overlapping records directly. Original HERON further filtered its training source internally, so the displayed percentage measures source exposure for that configuration. The diagnostic identifies cross-table partition alignment as a concrete source of evaluation bias. In the all-model comparison, HERON-plus had mean AUC/AP 0.8952/0.3916 (Figure 18; Table 15), exceeding m4 in 19 and 20 of the 20 shared folds, respectively (Figure 19). In the ten-fold head-to-head comparison, its AUC/AP rose from HERON's 0.8406/0.2686 to 0.8831/0.3624, with top-decile recall increasing from 48.97% to 59.48% (Table 16). Consistently high scores coexisted with extensive training exposure. Pairing test sets did not remove information sharing during fitting. Neural scores describe prediction with test exposure, and the geographic-transfer assessment therefore rests on the conventional models with separated training and testing. This information sharing weakens the use of neural scores to justify new monitoring locations and cannot strengthen a persistence interpretation for high-scoring environments. Lowland revisit candidates should instead be informed by the separated-training stratified results, so that coverage gains rest on identification of held-out reports. The metric comparison separates discrimination, probability error and coverage of highly ranked checklists (Figure 20). Conventional tree-model gains concentrate in ranking and selected probability-error measures, whereas the higher-dimensional MLP declines on multiple measures, consistent with its increased Brier score.

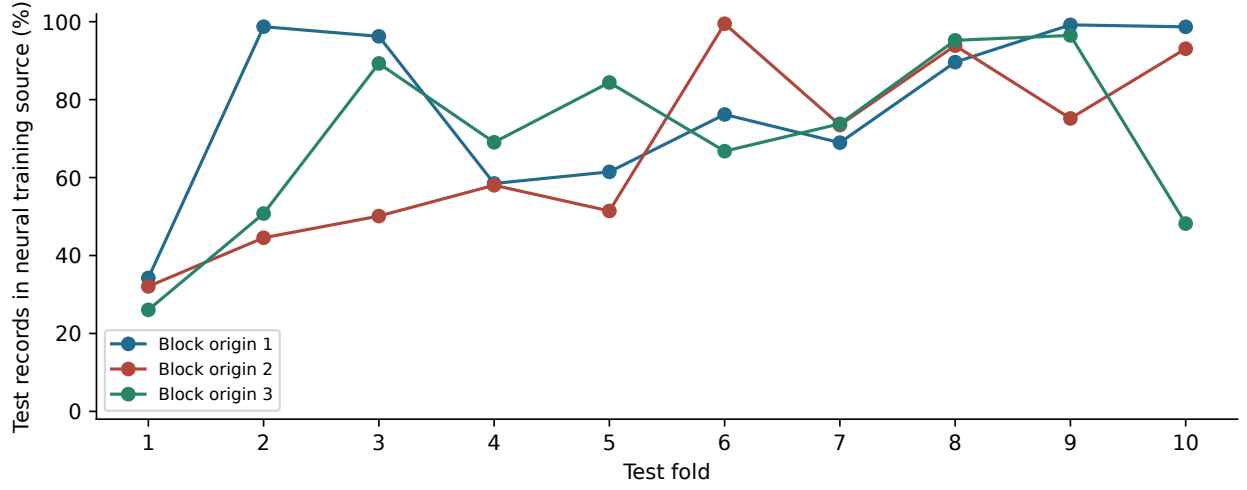


Figure 17: Fold-specific training-source overlap across three spatial partitions. Each line represents one partition and the horizontal axis identifies test folds. Text summaries weight all 68,976 test records within each partition.

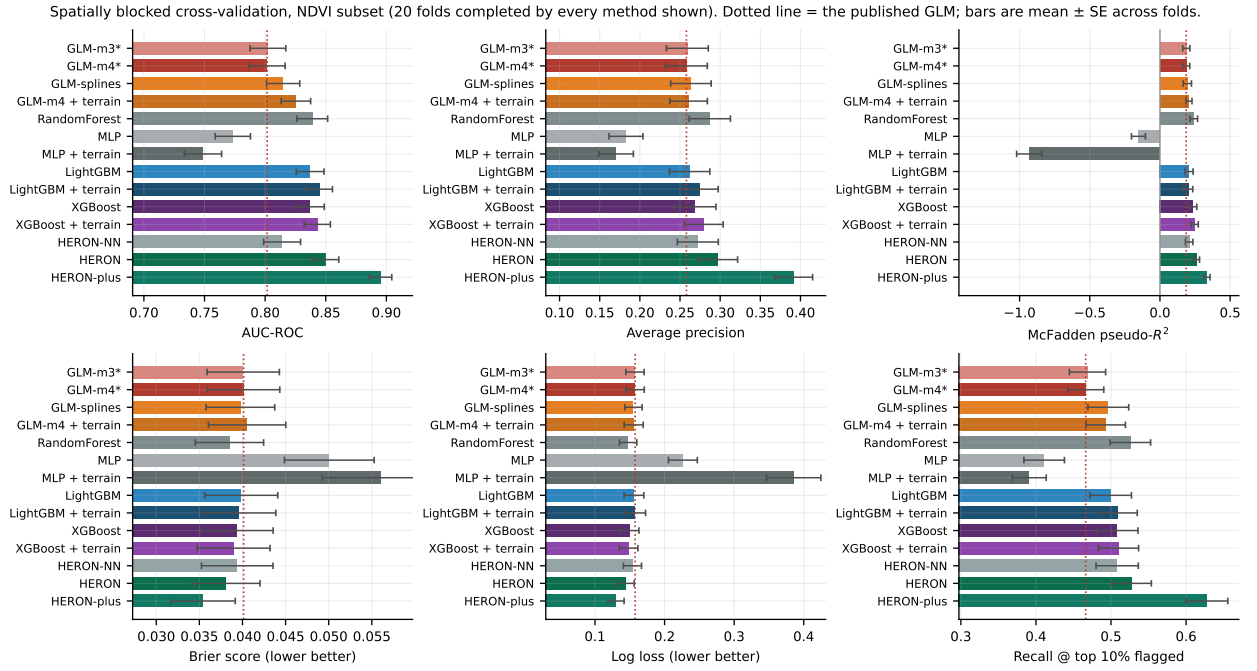


Figure 18: All-model performance on 20 common folds. Bars show standard errors across folds; neural evaluations include the training-source overlap in Figure 17.

Table 15: Fold means for the all-model spatial comparison: 30 folds for conventional models and 20 for neural configurations. Neural training-source overlap is shown in Figure 17.

Method	AUC	AP	pseudo- R^2	Tjur R^2	Brier	LogLoss	ECE	Recall at 10pct
	↑	↑	↑	↑	↓	↓	↓	↑
GLM-m3 (published)	0.8007	0.2540	0.1856	0.1244	0.0401	0.1573	0.0196	0.4624
GLM-m4 (published) ‡	0.8002	0.2530	0.1845	0.1240	0.0401	0.1575	0.0197	0.4614
GLM-splines	0.8119	0.2601	0.1948	0.1302	0.0397	0.1550	0.0208	0.4931
GLM-m4 + terrain	0.8217	0.2555	0.2040	0.1436	0.0403	0.1549	0.0206	0.4887
RandomForest	0.8423	0.2862	0.2418	0.1487	0.0384	0.1470	0.0181	0.5342
MLP	0.7771	0.1875	-0.1478	0.1634	0.0494	0.2245	0.0393	0.4177
MLP + terrain	0.7477	0.1675	-0.9572	0.1549	0.0564	0.3860	0.0512	0.3862
LightGBM	0.8391	0.2617	0.2068	0.1495	0.0397	0.1557	0.0190	0.5059
LightGBM + terrain	0.8461	0.2696	0.1962	0.1373	0.0396	0.1578	0.0209	0.5130
XGBoost	0.8397	0.2687	0.2355	0.1610	0.0392	0.1495	0.0155	0.5172
XGBoost + terrain	0.8438	0.2733	0.2408	0.1594	0.0389	0.1484	0.0169	0.5110
HERON-NN	0.8140	0.2722	0.2082	0.1298	0.0394	0.1538	0.0212	0.5082
HERON	0.8496	0.2972	0.2577	0.1538	0.0381	0.1442	0.0177	0.5274
HERON-plus	0.8952	0.3916	0.3341	0.2129	0.0354	0.1303	0.0187	0.6276

‡ the model published by [5].

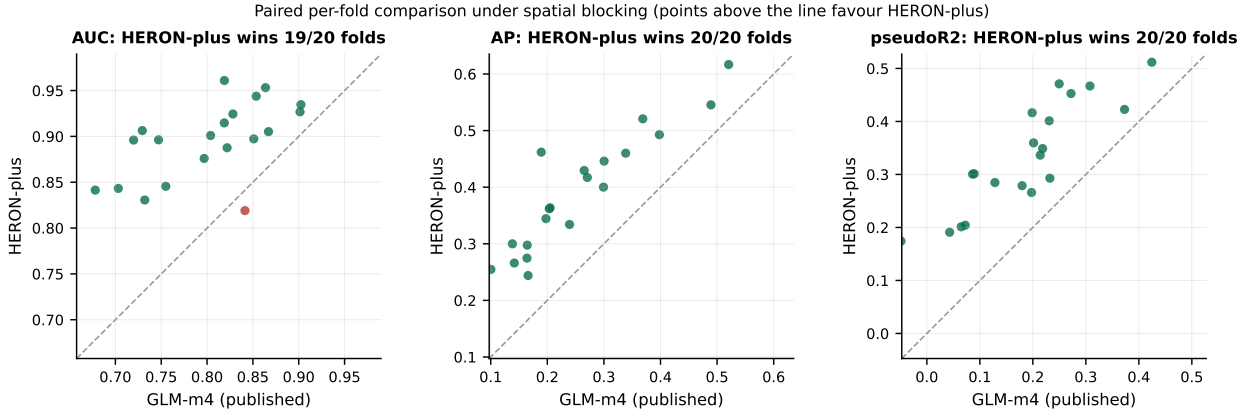


Figure 19: Paired fold results for HERON-plus and m4. Points represent shared test folds and the dashed line denotes equal scores; neural training sources overlap test observations.

Table 16: Ten-fold direct configuration comparison. Cross-table neural training-source overlap is analyzed in Section 3.13.

Method	AUC	AP	pseudo- R^2	Tjur R^2	Brier	LogLoss	ECE	Recall at 10pct
	↑	↑	↑	↑	↓	↓	↓	↑
GLM-m4 (published) ‡	0.7920	0.2337	0.1783	0.1092	0.0401	0.1576	0.0209	0.4361
HERON	0.8406	0.2686	0.2451	0.1331	0.0383	0.1451	0.0183	0.4897
HERON-plus	0.8831	0.3624	0.3202	0.1902	0.0351	0.1306	0.0175	0.5948

‡ the model published by [5].

F1, MCC, balanced accuracy and recall depend on the same predictions and threshold rule; they describe different uses of those predictions. For monitoring, the display identifies which tasks benefit and whether ranking gains accompany losses in probability quality. Higher recall can expand candidate records and better probability quality can support expected-report estimates. Both must be combined with environmental stratification to determine whether lowland coverage improves or upland tracking weakens. Agreement across metrics does not add observations of site-state transitions.

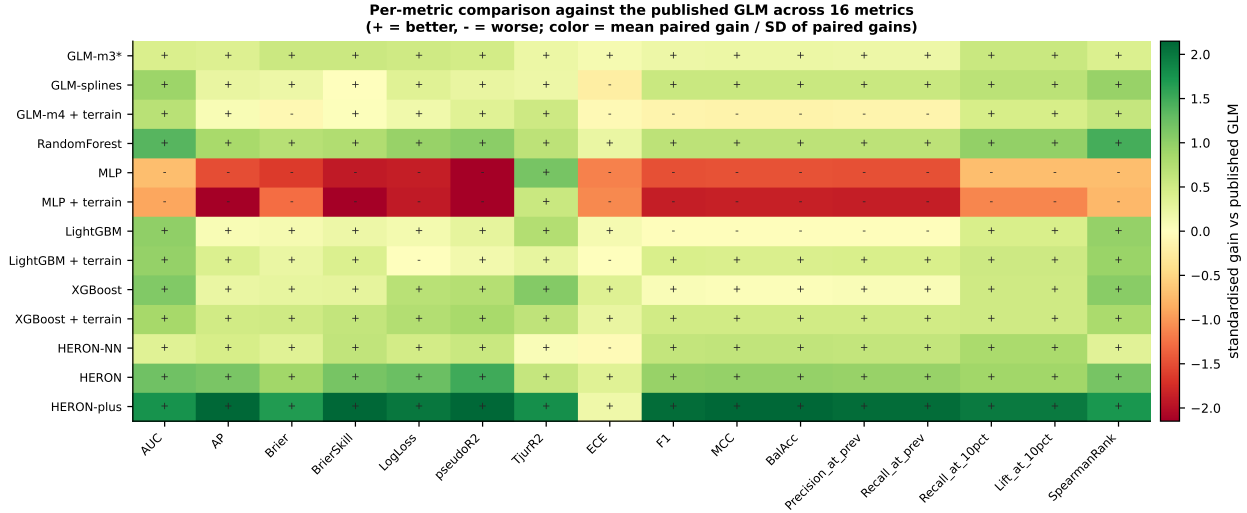


Figure 20: Differences from m4 across 16 prediction metrics. Colors scale gains by the standard deviation of fold differences; signs indicate the direction of improvement. No significance classification is applied.

3.14 Ablations highlight spatial inputs and tree members

Spatial Fourier features produced the largest neural-branch AUC increase (+0.0347), followed by terrain (+0.0204). Removing the grouped loss or spatial-risk penalty changed AUC by only -0.0002 in each case (Figure 21; Table 17). Increasing the grouped-loss weight fivefold reduced AP by 0.0124. Scores therefore responded mainly to spatial representation and environmental inputs, while grouped constraints supplied little ranking improvement. Removing monotone effort constraints increased AUC/AP by 0.0101/0.0078, showing the fitting restriction imposed by that structural choice. Platt calibration preserved AUC/AP but changed Brier from 0.03953 without calibration to 0.03971 with it, providing no improvement in overall probability error. These are internal responses of the exposed-training workflow; spatial features may also exploit that exposure. For environmental interpretation, gains from spatial representation cannot be assigned to stronger vegetation or elevation effects, and small grouped-loss responses add no direct persistence evidence. The ablation informs fitted record ranking; occupancy persistence still requires transition information from repeated surveys. In the six-fold stack ablation, removing tree members reduced AUC by 0.0311 and AP by 0.0271, whereas removing the neural member reduced them by only 0.0034 and 0.0046 (Table 18). Tree members therefore maintained most of the combination’s score, with a smaller neural increment. Removing the GLM increased AUC by 0.0010 but reduced AP by 0.0067 and increased Brier by 0.00036. The GLM contributed little to overall discrimination while supporting positive-report ranking and probability error. Stacking consequently reflected complementary contributions across metrics. For monitoring, this favors testing whether simpler tree models suffice for lowland candidate screening before assessing whether a combination adds coverage in particular environments. These member contributions were not decomposed by environment and do not establish that the neural branch located additional grids with persistent occupancy. Expanding the training source from 68,976 complete records to all 89,682 records increased six-fold stack AUC/AP from 0.8623/0.3343 to 0.8658/0.3398 (Figure 22). The AP increase of 0.0055 was smaller

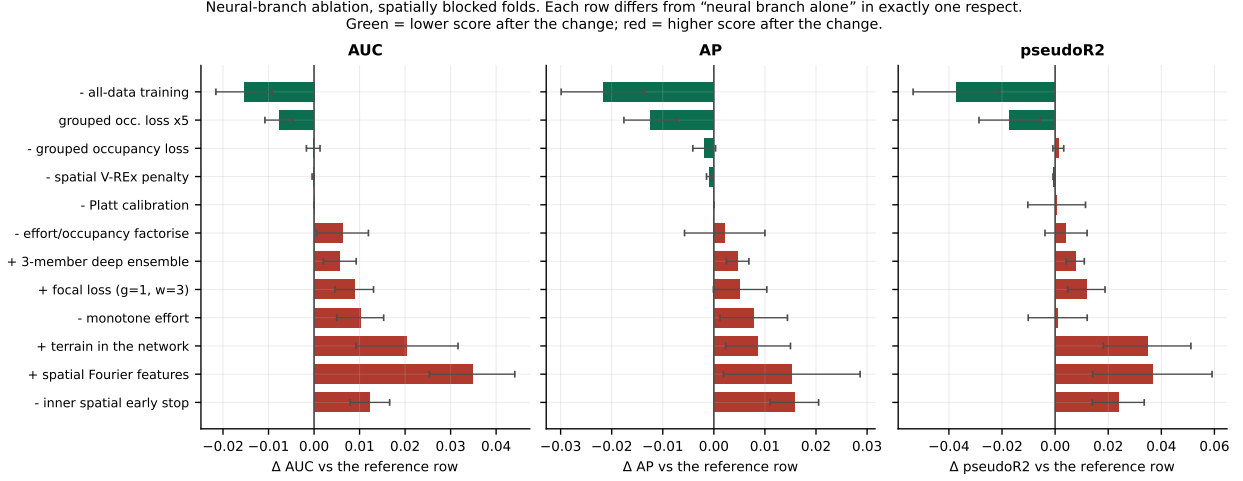


Figure 21: Twenty-fold neural-component comparisons. Deltas are relative to the base neural branch and bars show standard errors of paired differences; rows describe responses to component changes within this training workflow.

Table 17: Twenty-fold neural-branch ablation. Deltas use the base neural branch and retain each metric’s original direction.

Variant	AUC	Δ	AP	Δ	pseudo- R^2	Δ	Brier	Δ
neural branch alone	0.8084		0.2675		0.2006		0.03971	
- all-data training	0.7930	-0.0154	0.2458	-0.0217	0.1634	-0.0372	0.04104	+0.00134
grouped occ. loss x5	0.8007	-0.0077	0.2552	-0.0124	0.1834	-0.0172	0.04038	+0.00067
- grouped occupancy loss	0.8082	-0.0002	0.2656	-0.0019	0.2018	+0.0012	0.03972	+0.00001
- spatial V-REx penalty	0.8082	-0.0002	0.2667	-0.0008	0.2001	-0.0005	0.03975	+0.00004
- Platt calibration	0.8084	+0.0000	0.2675	+0.0000	0.2012	+0.0006	0.03953	-0.00018
- effort/occupancy factorise	0.8146	+0.0062	0.2697	+0.0021	0.2047	+0.0041	0.03923	-0.00048
+ 3-member deep ensemble	0.8140	+0.0056	0.2722	+0.0047	0.2082	+0.0076	0.03940	-0.00031
+ focal loss (g=1, w=3)	0.8172	+0.0088	0.2727	+0.0051	0.2123	+0.0118	0.03929	-0.00042
- monotone effort	0.8185	+0.0101	0.2753	+0.0078	0.2015	+0.0010	0.03949	-0.00021
+ terrain in the network	0.8288	+0.0204	0.2762	+0.0087	0.2352	+0.0347	0.03881	-0.00090
+ spatial Fourier features	0.8431	+0.0347	0.2828	+0.0153	0.2372	+0.0366	0.03829	-0.00142
- inner spatial early stop	0.8206	+0.0123	0.2833	+0.0158	0.2244	+0.0238	0.03865	-0.00106

Table 18: Six-fold stack ablation. Deltas use the full stack as reference.

Variant	AUC	Δ	AP	Δ	pseudo- R^2	Δ	Brier	Δ
HERON (full)	0.8658		0.3398		0.2631		0.03807	
- tree members	0.8347	-0.0311	0.3127	-0.0271	0.2258	-0.0373	0.03885	+0.00078
- GLM member	0.8668	+0.0010	0.3330	-0.0067	0.2573	-0.0058	0.03843	+0.00036
- all-data training	0.8623	-0.0035	0.3343	-0.0055	0.2576	-0.0054	0.03831	+0.00024
- neural member	0.8624	-0.0034	0.3351	-0.0046	0.2585	-0.0046	0.03826	+0.00019

than the 0.0217 contrast in the 20-fold neural-branch comparison, showing that additional records affected the branch and the combined predictor differently. Added records had a higher report rate (8.71% versus 4.89%), and expansion changed both report composition and cross-table training exposure. The score response therefore combines changes in data composition and partition alignment, rather than isolating environmental information supplied by the additional records. Given 49.6% NDVI missingness at high elevation, retaining missing records has a clear representational value for uplands, although these score increments do not isolate that benefit. Monitoring should preserve coverage of locations associated with missing records and recover comparable vegetation information before testing whether they change estimated upland reporting rebound and occupancy persistence.



Figure 22: All-record versus complete-record training in the six-fold stack. Lines connect results from the same fold; expansion changes both sample composition and training-source alignment.

3.15 Effort responses, environmental curves and centroids reveal fitted structure

The effort-branch score increased from 0.286 to 0.588 as duration rose from 5 to 240 minutes, and from 0.316 to 0.533 as distance rose from 0 to 5 km (Figure 23). Under otherwise fixed conditions, more observation effort therefore yields a higher report prediction. This illustrates the role of effort standardization in period comparisons. Monotonic constraints set the response direction, while fitting determines slope and curvature; the curves explain how effort adjusts the reporting score. Differences in visit duration across elevations or periods could amplify or weaken unstandardized reporting contrasts. Comparable duration and distance, together with retained nonreporting visits, allow repeat surveys and detection models to test whether increased reporting accompanies higher occupancy persistence. Elevation responses were nonlinear. The neural environmental branch approached saturation at higher elevations, whereas the GLM's linear-logit response continued rising (Figure 24). The NDVI response was unimodal, with reporting-response peaks near 0.53, 0.47 and 0.51 across the three periods. Flexible fitting thus separated the greenest conditions from the highest fitted score. Together with the GLM's average positive association, these curves distinguish an overall gradient from local curvature and identify high-greenness sample coverage and covariate combinations as relevant checks on the fitted peak. For survey placement, this model disagreement supports retaining both intermediate- and high-greenness sites to test whether the reporting curvature appears in repeat observations. Training exposure and the branch-score definition weaken an interpretation of the peak as optimal greenness for persistence, providing no basis to reduce revisits in the greenest environments. Response shape explains the different absolute elevation centroids (Figure 25). GLM centroids were 939.7, 968.2 and 953.5 m, versus 797.8, 838.7 and 817.4 m for the neural reporting curve. High-elevation saturation reduced the neural curve's relative weight at the upper end of the prediction grid, lowering its absolute centroid. Both models showed an upward centroid shift between hurricanes followed by a partial return. Shifts relative to the initial period were +28.5 and +13.9 m for the GLM, and +40.9 and +19.6 m for the neural model. This shared

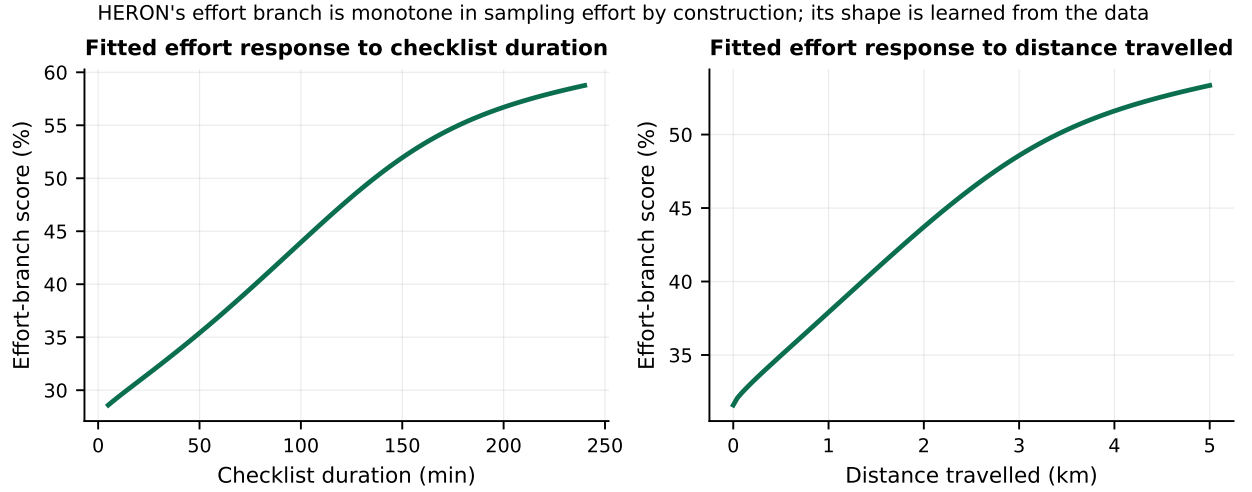


Figure 23: Effort-branch responses with other inputs fixed. Monotonic constraints require nondecreasing scores with duration and distance.

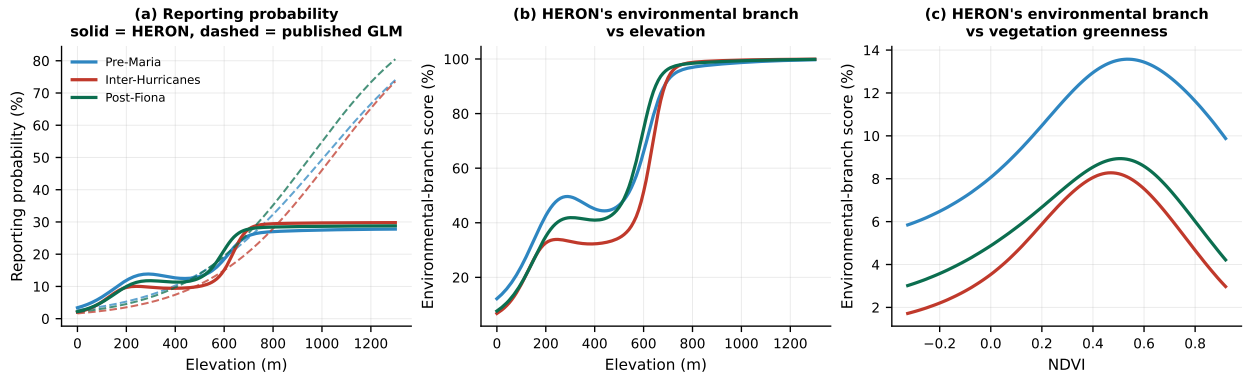
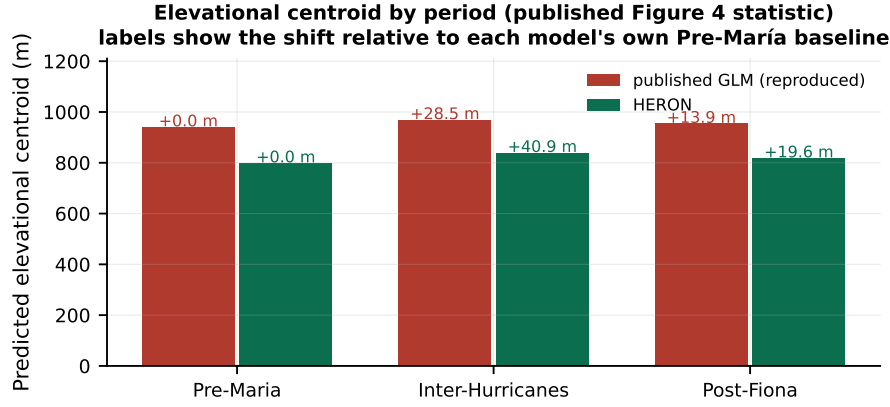


Figure 24: GLM reporting curves and neural reporting and environmental-branch responses. Solid lines denote neural fits and dashed lines denote the GLM; branch scores use fixed conditional inputs.

ordering summarizes period-specific redistribution of fitted reporting weights, while its magnitude depends on response shape. Period comparisons therefore require both lowland and upland coverage. Tracking only high-reporting upland locations could miss record changes associated with the later downward reweighting. The curves share observations and their common ordering is not independent validation; same-site detection histories and transition models can test whether the redistribution accompanies persistent lowland occupancy.



Absolute heights are not comparable between model classes: the GLM's linear logit keeps rising to 1300 m while HERON saturates above ~700 m. The within-model shifts are the comparable quantity.

Figure 25: Probability-weighted elevation centroids on the 0–1300 m prediction grid. Annotations show shifts relative to each model's own initial baseline.

3.16 Spatial heterogeneity and variable contributions inform monitoring

In the reproduced original occupancy model, predicted local extinction at mean greenness declined from 56.7% near sea level to 0.9% at approximately 1300 m (Figure 26). The broad upper-elevation interval indicates limited information about the magnitude at environmental extremes. The negative elevation coefficient after observation-order correction retained the same gradient direction. The complement of the neural environmental branch also declined with elevation, but mathematically expresses a low environmental score. The panels thus show a site-transition estimate and a conditional score response; the persistence interpretation derives from the dynamic model's transition component. The elevation gradient supplies a process-based rationale for fixed upland revisits, while higher modeled lowland transition risk motivates lowland greenness contrasts. Together with higher corrected persistence at greener lowland sites, it supports testing which lowland conditions are associated with higher occupancy persistence probabilities rather than treating all lowlands as high-loss environments. Site maps translate these gradients into monitoring-placement informa-

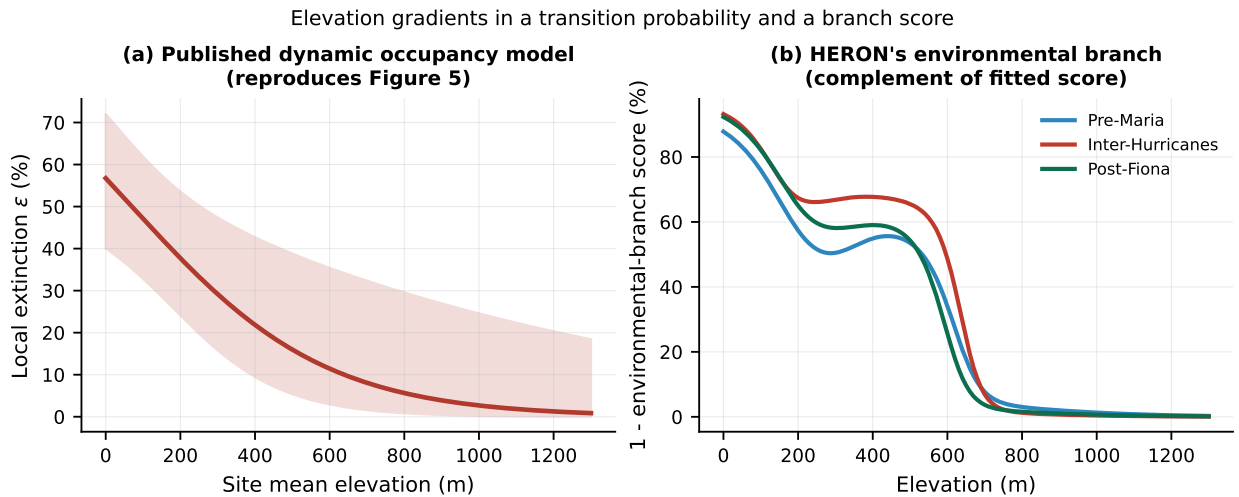


Figure 26: Two model outputs along elevation. (a) Reproduced original dynamic-model local extinction probability and interval; (b) one minus the neural environmental-branch score.

tion (Figure 27). Transition predictions at occupancy grids and neural scores at 3,727 sampled sites both vary spatially, while observation locations are unevenly distributed. The maps identify score contrasts and sampling gaps within the observed domain, supporting strata that include both

upland and lowland settings. The neural map uses cross-period mean site NDVI and therefore expresses period scores under fixed site conditions; time-varying vegetation recovery requires concurrent remote-sensing and repeated bird observations. The maps can guide selection of revisit sites across environments. Repeat surveys in upland grids with high modeled occupancy persistence can track occupancy transitions, while lowland greenness contrasts can test whether incomplete reporting rebound coexists with high occupancy persistence, and observation gaps identify coverage needs. Candidate locations still require assessment of access and survey conditions; mapped scores themselves do not demonstrate site recovery. Permutation importance exposed different information

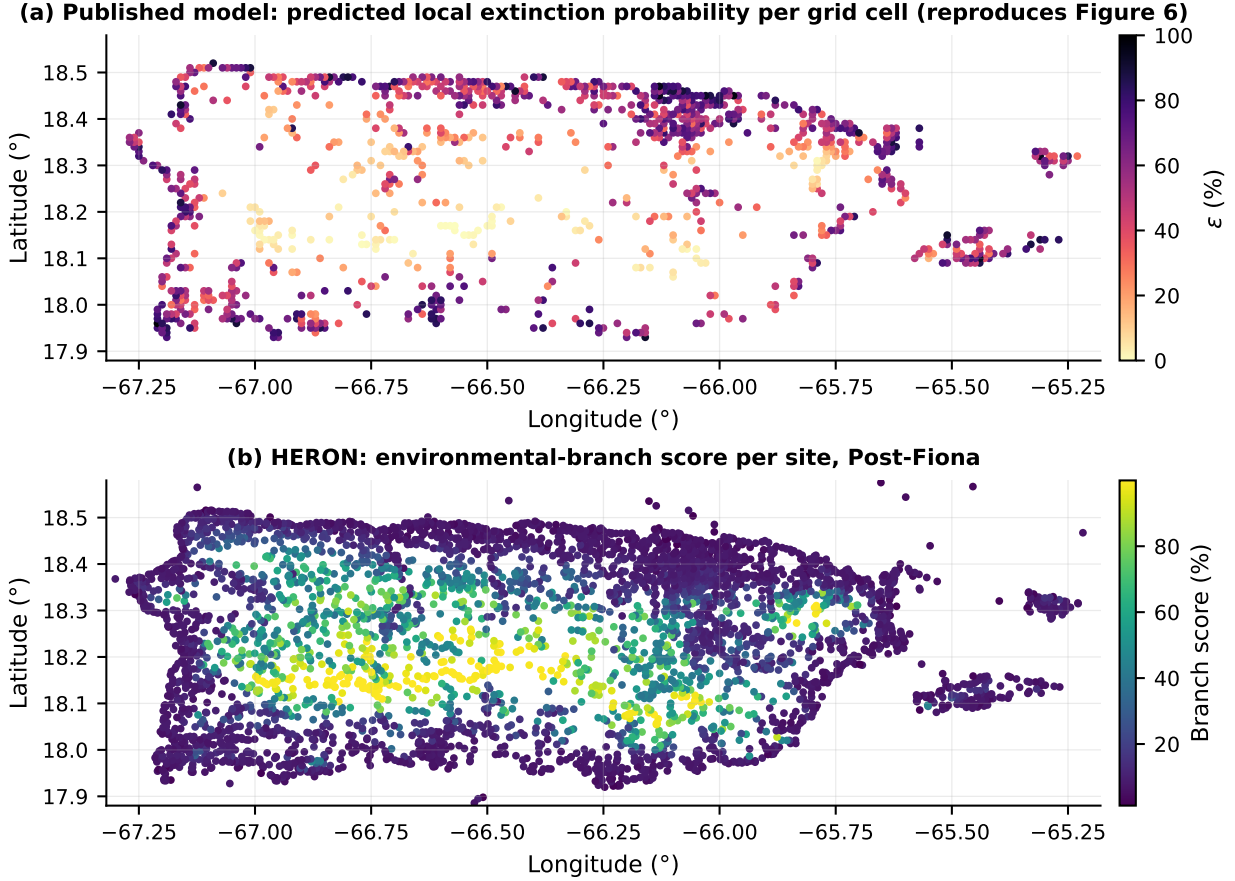


Figure 27: Site-level spatial heterogeneity. The upper panel shows reproduced original dynamic-model local extinction predictions. The lower panel shows neural environmental-branch scores at sampled sites in the Post-Fiona period.

dependencies (Figure 28). Permuting elevation reduced neural-branch AP by 0.0809, versus 0.0075 for greenness, identifying elevation as the branch's main organizing gradient. In the held-out terrain-LightGBM fold, duration contributed most (AP reduction 0.0898), followed by longitude (0.0354), broad-scale mean slope (0.0246) and latitude (0.0212). Tree-model predictions therefore depended on a combination of observation effort, geographic position and multiscale terrain. Recording effort consistently while preserving spatial coverage helps separate reporting differences associated with observation opportunity from those associated with site conditions.

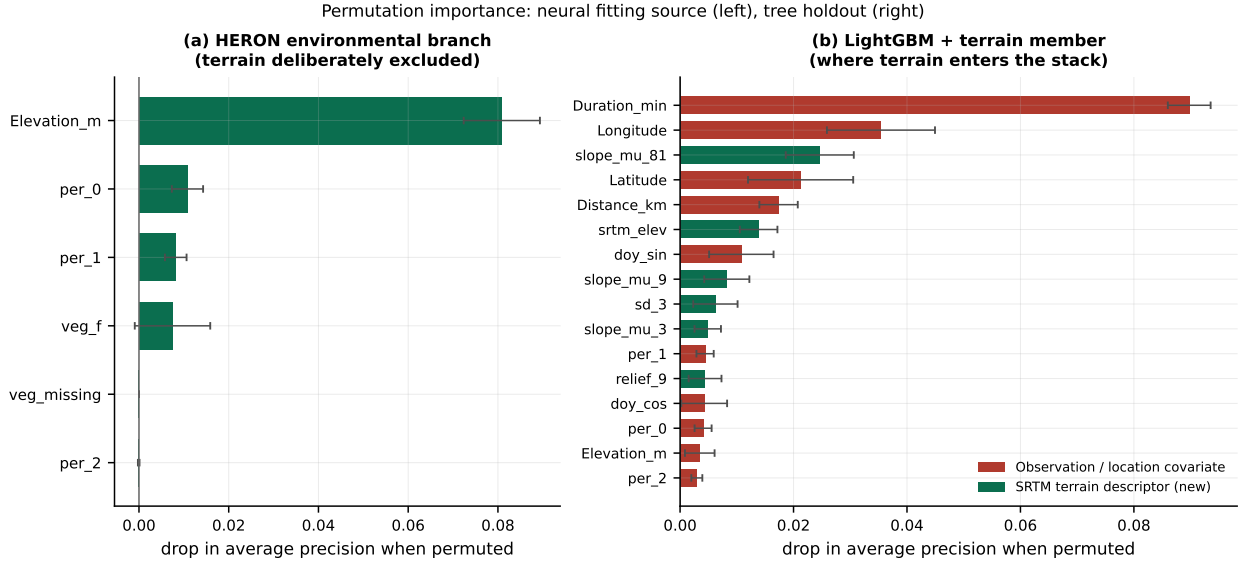


Figure 28: AP reductions after feature permutation, with standard deviations across permutations. The neural panel uses fitting-source observations and the LightGBM panel uses one geographic holdout; each describes its model’s information dependence.

4 Discussion

4.1 Larger rebound point estimates occur in high, green environments

The environmental meaning of rebound comes from within-stratum changes rather than the island-wide mean alone. Between Inter-Hurricanes and Post-Fiona, low-elevation low- and high-greenness cells increased by 0.39 and 1.07 percentage points, compared with 6.97 points in the high-elevation, high-greenness cell. The latter also had a high initial reporting level, so its larger absolute increment must be interpreted against that baseline. Both lowland cells retained later-period point estimates below their initial values, suggesting that rebound did not take the form of a uniform return to earlier reporting across environments. Spatially clustered intervals for every Post-Fiona versus initial contrast span zero. The evidence establishes environmental differences in reporting levels more clearly than a categorical separation into recovered and unrecovered environments.

This interpretation also depends on which environments enter the analysis. High-elevation NDVI missingness reached 49.6%, and high-elevation checklists constituted 11.2% of missing records but only 3.4% of complete records. The missing-record reporting odds ratio declined from 1.86 to 1.03 after elevation and period adjustment, identifying an environmental source of aggregate reporting differences. Standardization improves period comparability by holding within-cell covariate distributions constant, while recovering vegetation information at missing-record locations would broaden environmental coverage. Together, these steps help locate rebound without treating changes in sampling weights as changes shared by all sites. This concern is consistent with the interacting sampling biases identified by Backstrom et al. [8]; their results support checking environmental representation alongside survey effort.

Elevation and greenness associations are compatible with the ecological context of forest damage and bird responses following Puerto Rican hurricanes [1, 2]. Howe et al. [4] used LiDAR and NDVI to describe complementary structural and greenness responses in mangroves. Their distinction matters here because NDVI alone cannot establish canopy recovery or floral-resource availability. Rivera-Milán et al. [3] found prolonged post-hurricane depletion in the Puerto Rico Plain Pigeon, further showing that a common disturbance does not imply a common recovery trajectory across bird taxa. These comparisons provide ecological context rather than independent confirmation of the Emerald pattern. Elevation also represents climate, habitat and accessibility, and the three periods combine two storms and elapsed time without an undisturbed counterfactual, preventing attribution of the period contrast specifically to Fiona.

4.2 High reporting and persistence coincide in uplands but require distinction within lowlands

High-elevation, high-greenness environments combine high reporting with high occupancy persistence probability, giving directionally consistent environmental patterns. Correcting observation alignment retained the associations of elevation and greenness with lower local extinction while reducing the absolute initial-period detection coefficient by approximately 90%. This retains support for linking high reporting in higher, greener settings with occupancy persistence and weakens the earlier strong period-detection contrast. Because the reporting and occupancy models share observations, their agreement compares different response definitions rather than providing independent biological validation.

Lowland results refine this interpretation. The later reporting prediction for the low-elevation, high-greenness cell was 5.17%, below its initial 6.27%, yet mean occupancy persistence probability at corresponding sites was 71.3%, compared with 24.2% in the low-elevation, low-greenness cell. Incomplete reporting rebound can therefore coexist with relatively strong occupancy persistence probability. The former describes checklist-level reporting trajectories and the latter retention of previously occupied sites. Monitoring must consequently revisit both previously reporting and previously nonreporting sites, estimating state transitions under imperfect detection. Treating all lowlands as sharing a recovery state would obscure their greenness-related differences.

At a fixed elevation of 100 m, changing NDVI from 0.30 to 0.76 corresponded to a 42.2-point persistence contrast but only a 0.7-point colonization contrast with an interval spanning zero. The model supplies clearer support for greenness-associated retention than for greenness-associated entry into occupancy. Under equal visit counts, repeat reports occurred in 1/9 and 15/28 low- and high-greenness lowland pairs, providing a directionally similar observation pattern, although these small-sample histories combine detection and state changes. Greener lowlands therefore merit targeted occupancy surveys, with less-green sites retained as contrasts to distinguish lower occupancy probabilities from inadequate detection.

These probabilities describe between-site associations at the model's scale. Cross-period mean NDVI does not establish the temporal sequence of vegetation recovery within sites, and repeat visits within multi-year primary periods rely on closure. The results identify where to examine persistence; determining whether vegetation change improves persistence requires simultaneous state and vegetation observations within shorter, defensible survey windows. Gordon et al. [12] linked colonization and extinction to time since management disturbance using repeated point counts. For the Emerald, recording disturbance and vegetation histories at fixed grids would similarly allow habitat change to be related to separate transition processes.

4.3 Monitoring should retain upland persistence sites and lowland environmental contrasts

Only 3.3% of lowland checklists reported the species, yet lowlands contained 60.7% of complete-sample positive reports. Lowlands are therefore both difficult reporting environments and an indispensable part of persistence monitoring. Random forest improved AP at low and middle elevations without a comparable high-elevation gain, locating its value in finding sparse reports among many nonreports. Within the global top decile, lowland positive coverage increased from m4's 26.4% to 34.6%, while upland coverage declined. Better prediction does not automatically yield more balanced environmental monitoring; ranking must serve the intended environmental contrasts.

Equal-budget comparison quantifies this choice. Global random-forest ranking selected 57.9% of high-elevation checklists but 6.9% of low-elevation checklists. Allocating approximately 10% within each band increased lowland positive coverage to 45.6% while capturing 388 fewer positives overall. Global ranking favors aggregate hits, whereas comparing occupancy persistence across lowland conditions requires explicit lowland allocation. This comparison concerns historical checklist screening, not the number of independent survey sites, actual revisit costs or improved precision of persistence estimates.

Monitoring can therefore combine three complementary components. First, fixed upland sites can supply detection histories for estimating occupancy transitions in high-reporting environments, with added coverage of sparse conditions such as less-green uplands. Second, lowland coverage should include low, middle and high greenness to compare reporting changes and occupancy persistence across greenness conditions. Third, spatially evaluated ranking can identify supplementary candidates within these strata while fixed sites and nonreporting sites remain in the survey. Candidate screening would then complement complete state-transition observations. Repeat visits should record duration, distance, concurrent vegetation and available floral-resource information to distinguish environmental change from detection opportunity. The fixed routes and interval recordings in the bird-monitoring system of Ovaskainen et al. [19] provide a recent example of stabilizing observation conditions alongside predictive modelling. Here, that principle supports maintaining comparable repeat visits even when model rankings change.

4.4 Spatial and temporal validation delimit the use of revisit clues

Prediction experiments support these monitoring roles to different degrees across applications. Random forest’s AP advantage over m4 narrowed from 0.143 under random splitting to 0.033 under spatial splitting. High scores within familiar location mixtures therefore overstate the corresponding gain in held-out geographic blocks. Increasing block width further reduced some flexible-model advantages. Spatial validation helps assess the domain over which candidate ranking remains informative and supports retaining environmental coverage as surveys expand [13–15]. The current island-internal evaluation without distance buffers does not establish performance in every unsampled environment. Koldasbayeva and Zaytsev [16] likewise emphasize blocked evaluation and out-of-time testing for distribution prediction. Their results motivate matching validation to the intended revisit domain; they do not establish that the block sizes used here remove all spatial dependence.

In the Post-Fiona temporal holdout, random forest increased top-decile recall from m4’s 53.28% to 59.80%, showing that earlier relationships still helped identify later reports. This supports using existing records to locate follow-up opportunities but does not separately establish temporal gains in each lowland greenness cell. Spatial stratification identifies the environments that benefit, whereas temporal holdout assesses whether aggregate reporting information carries into a later period. Their joint interpretation helps plan revisits without replacing repeated same-site detection records and occupancy-transition analysis. Lovell et al. [17] found that spatial and temporal environmental effects on occupancy and abundance differed across parts of a butterfly’s range. This offers a methodological parallel for testing temporal performance within environmental strata rather than assuming that an overall holdout gain applies uniformly.

The remaining experiments specify how prediction tools serve the environmental question. EVI2 results show that ranking gains are not confined to NDVI; calibration separates candidate ranking from expected-report estimation; effort responses and importance analyses emphasize comparable survey conditions. Similar spatial performance of m3 and m4 shows that an interaction describing period–elevation patterns need not improve new-site ranking, reinforcing separate evidential requirements for explanation and prediction [18]. Neural training-source overlap weakens the use of its high scores to justify new monitoring locations. Its curves and ablations describe fitted information dependencies but do not strengthen process conclusions about persistence. Maps and curves should ultimately identify environmental contrasts for revisits, with repeated surveys and detection models estimating whether occupied grids remain occupied in later periods.

5 Conclusions

Puerto Rican Emerald reporting rebound is structured by elevation and greenness. High-elevation, high-greenness environments combine larger absolute rebound point estimates with high reporting levels, while recovery relative to initial levels within each stratum requires further repeated observation. These environments also have high modeled persistence. Greener lowlands retain a stronger

persistence association, showing that incomplete lowland reporting rebound cannot be equated with nonoccupancy at all previously occupied grids. Prediction gains chiefly identify sparse low- and middle-elevation reports, and spatial and temporal validation delimit their use as revisit clues. Monitoring should combine fixed upland repeat sites, lowland greenness contrasts and within-stratum candidate screening to examine reporting changes and repeat records separately from occupancy persistence and colonization estimated under imperfect detection.

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